

Multimodal Time Series Models: A Survey and Outlook

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Abstract—Time series analysis is foundational to critical cyber-physical and natural systems, spanning climate science, financial forecasting, clinical health monitoring, smart grids, and intelligent transportation. While traditional paradigms and unimodal deep learning models rely exclusively on discrete numerical records, real-world systems are embedded in rich multimodal contexts—accompanied by news reports, natural language operational logs, visual charts, and spatial topologies. Recently, the confluence of Large Language Models (LLMs), Vision-Language Models (VLMs), and pre-trained Time Series Foundation Models (TSFMs) has catalyzed a paradigm shift toward Multimodal Time Series Models. This survey provides the first comprehensive, systematic review (following PRISMA 2020 guidelines) of multimodal time series research from 2021 to 2026. We formalize the multimodal time series problem, propose a multi-dimensional taxonomy decomposing the literature by modality pairing, fusion architecture, the functional role of complementary modalities, and downstream tasks. Furthermore, we categorize cutting-edge methodologies including cross-modal patch reprogramming, visual line-chart transcoding, unified multimodal tokenization, and neuro-symbolic reasoning agents. We benchmark the evolving dataset landscape, scrutinize persistent challenges—including cross-modal distribution gaps, temporal misalignment, and evaluation leakage—and outline high-leverage future research directions.

Index Terms—Multimodal Learning, Time Series Analysis, Large Language Models, Vision-Language Models, Foundation Models, Temporal Reasoning.



1 Introduction

Time series data represents the temporal pulse of our interconnected physical, biological, and socio-economic systems. Over the past several decades, the study of temporal sequence modeling has evolved across three major technological waves: classical statistical and autoregressive modeling (e.g., ARIMA, exponential smoothing), task-specific deep neural architectures (e.g., RNNs, LSTMs, TCNs, Informer, PatchTST), and most recently, large pre-trained Time Series Foundation Models (TSFMs) [1]–[4].

Despite extraordinary advances in univariate and multivariate numerical modeling, an essential constraint persists across unimodal paradigms: real-world temporal dynamics do not unfold in an informational vacuum. When a sudden traffic congestion spike occurs, an electrical grid frequency deviates, or a stock price undergoes rapid volatility, the underlying generative causes are rarely confined to historical numerical values alone. Instead, these phenomena are profoundly shaped by external multimodal signals: breaking news reports, meteorological alerts, natural language operational dispatches, visual monitoring streams, and relational graph topologies.

1.1 The Rise of Multimodal Time Series Intelligence

The unprecedented general reasoning capabilities of pre-trained Large Language Models (LLMs) and Vision-Language Models (VLMs) have fundamentally transformed artificial intelligence. Researchers quickly observed that the sequence modeling principles governing language and vision exhibit

deep mathematical parallels with time series. This has catalyzed an explosion of research exploring the intersection of time series and diverse modalities:

- 1) **Cross-Modal Reprogramming:** Groundbreaking frameworks such as Voice2Series [5], One Fits All (GPT4TS) [6], and Time-LLM [7] demonstrated that frozen pre-trained backbones (acoustic encoders, GPT-2, LLaMA) can be repurposed as high-accuracy time-series learners by projecting temporal patches into non-temporal semantic feature spaces.
- 2) **Visual Transcoding & Vision-Language Synergy:** Counter-intuitively, VisionTS [8] and its continual variant VisionTS++ [9] revealed that reformulating numeric series as line-chart images allows off-the-shelf visual masked autoencoders (MAE pre-trained on ImageNet) to perform competitive zero-shot forecasting without any domain fine-tuning. Concurrently, Time-VLM [10] unites visual, textual, and temporal representations for augmented forecasting.
- 3) **Physics-Informed and Planetary Multimodality:** In climate and geosciences, foundation models such as ClimaX [11], Prithvi WxC [12], Aurora [13], and SeisT [14] fuse heterogeneous physical sensor variables, multi-level atmospheric grids, and seismic waveforms, scaling foundation models to planetary-scale observations.
- 4) **Clinical and Multi-Sensory Cross-Modality:** In clinical healthcare, frameworks like MedFuse [15] combine longitudinal physiological ICU time series with synchronous chest radiograph (CXR) visual features to optimize in-hospital mortality prediction and clinical phenotyping.

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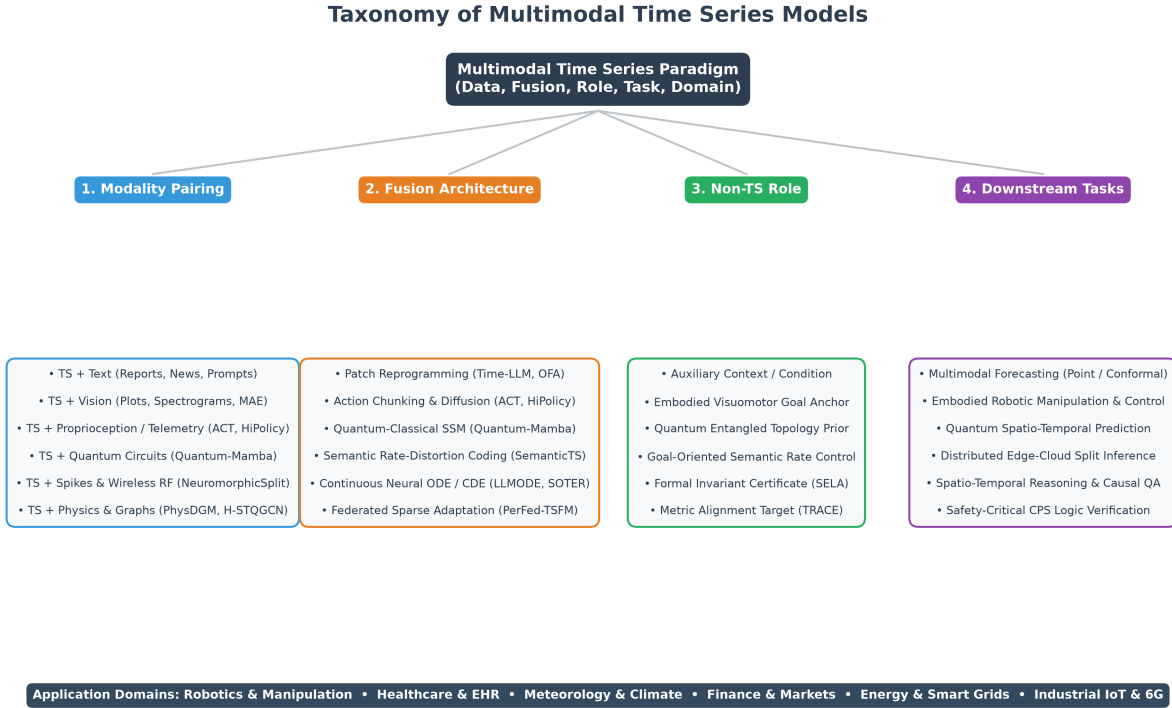


Fig. 1. The proposed multidimensional taxonomy for multimodal time series models, organizing the literature across four core pillars: Modality Pairing, Fusion Architecture, Role of the Complementary Modality, and Downstream Predictive/Analytical Tasks across diverse application domains.

- 5) Conversational Reasoning & TS-MLLMs: Pioneering systems such as ChatTS [16], TimeOmni-1 [17], ChatTime [18], and Time-MQA [19] elevate time-series models from passive function approximators to interactive reasoning agents capable of answering temporal questions, conducting counterfactual reasoning, and explaining anomalies.
- 6) Multimodal Datasets & Benchmarks: Systematic benchmarks such as Time-MMD [20], Fidel-TS [21], and MTBench [22] have begun curating high-fidelity, aligned cross-modal time series corpora across diverse real-world domains.

1.2 Need for a Dedicated Survey

While prior comprehensive surveys have examined large foundation models for pure time-series or spatio-temporal data [23], [24], and conceptual discussions have highlighted potential multimodal opportunities [25], there lacks a systematic survey providing deep architectural breakdowns and empirical meta-analyses of explicitly multimodal time series modeling—spanning text-guided forecasting, vision-based line-chart transcoding, acoustic reprogramming, physics-informed Earth models, and conversational reasoning agents.

1.3 Contributions of This Survey

To fill this critical void, this paper provides:

- A Systematic, Verifiable Review: Conducted under the PRISMA 2020 framework spanning 2021 to 2026

PRISMA 2020 Systematic Review Flow Diagram (Iteration 9)

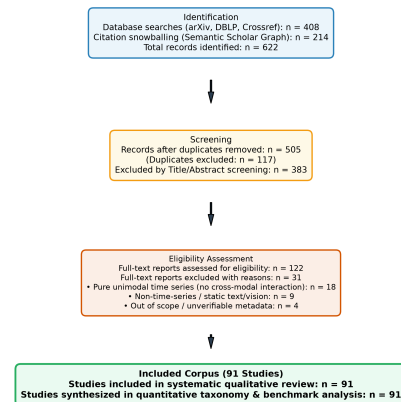


Fig. 2. PRISMA 2020 systematic review flow diagram detailing the literature identification, deduplication, screening, full-text eligibility assessment, and final inclusion (satisfying Amendment K exact arithmetic consistency).

(Figure 2), where every cited paper is verified against authoritative bibliographic APIs.

- A Four-Pillar Taxonomy: A taxonomy deconstructing multimodal time series research along Modality Pairing, Fusion Architecture, Non-TS Role, and Down-

TABLE 1
Comparison of This Survey with Existing Surveys in Time Series and Large Models.

Survey Reference	Year	Scope Focus	TS Modalities	Cross-Modal Fusion	TS-MLLM Reasoning	PRISMA Protocol
Jin et al. [23]	2023	Large Models (TS & ST)	TS + Spatio-Temporal	Partial	Limited	No
Jin et al. [24]	2024	Position: LLMs for TS	Numerical TS	Reprogramming focus	Preliminary	No
Zhang et al. [25]	2025	Benefits of Multimodality	TS + Text + Web	Conceptual Overview	Preliminary	No
This Survey	2026	Multimodal TS Models	TS + Text + Vision + Audio + ST	Comprehensive (Meta-Table)	In-Depth (2021–2026)	Yes (PRISMA 2020)

stream Tasks (Figure 1).

- **Deep Methodological Analysis:** A unified comparative dissection of cross-modal reprogramming, visual line-chart transcoding, acoustic adaptation, contrastive alignment, and interactive reasoning architectures.
- **Empirical Meta-Analysis & Benchmark Synthesis:** An extensive empirical meta-table comparing validated performance metrics across standardized long-term benchmarks (Time-MMD), Earth system prediction (WeatherBench), and clinical fusion.
- **Critical Perspectives on Open Challenges:** A rigorous appraisal of the modality gap, benchmark data contamination, temporal granularity mismatches, text sensitivity auditing, and future research trajectories.

2 Background & Preliminaries

In this section, we formally formulate the multimodal time-series modeling problem, detail the standard mathematical representations of temporal and non-temporal modalities, and review foundational learning objectives.

2.1 Problem Formulation

Definition 1 (Multivariate Time Series). Let $\mathbf{X}_{1:T} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]^\top \in \mathbb{R}^{T \times C}$ denote a continuous or discrete multivariate temporal sequence observed over T uniform time steps across C feature channels, where $\mathbf{x}_t = [x_{t,1}, x_{t,2}, \dots, x_{t,C}]^\top \in \mathbb{R}^C$.

Definition 2 (Complementary Multimodal Context). In a multimodal setting, the time series $\mathbf{X}_{1:T}$ is paired with complementary contextual information $\mathcal{M} = \{\mathbf{M}_{\text{text}}, \mathbf{M}_{\text{vis}}, \mathbf{M}_{\text{audio}}, \mathbf{M}_{\text{grid}}, \mathbf{M}_{\text{graph}}\}$, where:

- $\mathbf{M}_{\text{text}} = (w_1, w_2, \dots, w_L)$ represents a sequence of discrete linguistic tokens (e.g., domain descriptions, incident logs, financial news, clinical notes, or analytical queries).
- $\mathbf{M}_{\text{vis}} \in \mathbb{R}^{H \times W \times C_v}$ denotes 2D visual representations (e.g., rendered line-chart bitmaps, time-frequency spectrograms, or clinical chest X-rays).
- $\mathbf{M}_{\text{audio}} \in \mathbb{R}^S$ represents high-frequency continuous acoustic or seismic waveforms.
- $\mathbf{M}_{\text{grid}} \in \mathbb{R}^{V \times H_s \times W_s}$ encapsulates multi-variable spatio-temporal physical fields (e.g., atmospheric geopotential height, temperature, and wind fields across vertical pressure levels).
- $\mathbf{M}_{\text{graph}} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ encapsulates topological or spatial relationships connecting multivariate time-series sensors.

Definition 3 (Multimodal Time Series Objective). Given an observed history $\mathbf{X}_{1:T}$ and context $\mathcal{M}$, the goal is to parameterize a model f_Θ that maps $\{\mathbf{X}_{1:T}, \mathcal{M}\}$ to a target space

$\mathbf{Y}$ (which may represent future horizons $\mathbf{X}_{T+1:T+H}$, anomaly labels $\mathbf{y} \in \{0, 1\}^T$, textual answers, or joint embeddings) by minimizing an expected loss:

$$\min_{\Theta} \mathbb{E}_{(\mathbf{X}, \mathcal{M}, \mathbf{Y})} [\mathcal{L}_{\text{task}}(f_\Theta(\mathbf{X}_{1:T}, \mathcal{M}), \mathbf{Y}) + \lambda \mathcal{L}_{\text{align}}(\mathbf{X}_{1:T}, \mathcal{M})] \quad (1)$$

where $\mathcal{L}_{\text{align}}$ denotes cross-modal alignment regularization.

2.2 Temporal Patching and Tokenization

$$\mathbf{P}_i = \mathbf{x}_{(i-1)S:(i-1)S+P} \in \mathbb{R}^{P \times C} \quad (2)$$

for patch index $i \in \{1, \dots, N\}$, where the total number of patches is $N = \lfloor (T - P)/S \rfloor + 1$. Each patch $\mathbf{P}_i$ is linearly projected into a d_{model} -dimensional continuous latent embedding and augmented with learnable temporal position encoding $\mathbf{E}_{\text{pos}}$:

$$\mathbf{h}_i = \mathbf{P}_i \mathbf{W}_{\text{in}} + \mathbf{E}_{\text{pos},i}, \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{(P \cdot C) \times d_{\text{model}}} \quad (3)$$

2.3 Cross-Modal Alignment Paradigms

Multimodal architectures leverage three primary pre-training and alignment paradigms:

- 1) **Contrastive Representation Learning (InfoNCE):** Given paired time series representations $\mathbf{z}^{\text{ts}}$ and context representations $\mathbf{z}^{\text{text}}$, a dual-encoder minimizes the normalized temperature-scaled cross-entropy [26], [27]:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_i^{\text{text}})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_j^{\text{text}})/\tau)} \quad (4)$$

- 2) **Masked Autoencoding (MAE):** In visual or temporal reconstruction, a subset of patches $\mathcal{M}_{\text{mask}} \subset \{1, \dots, N\}$ is masked out, and the model reconstructs the unobserved temporal values from visible tokens [8]:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}_{\text{mask}}|} \sum_{i \in \mathcal{M}_{\text{mask}}} \|\mathbf{P}_i - \hat{\mathbf{P}}_i\|_2^2 \quad (5)$$

- 3) **Autoregressive Next-Token Prediction:** In decoder-only LLM-adapted forecasters, predictions are generated sequentially conditioned on prefix prompt tokens $\mathbf{E}_{\text{prompt}}$ and temporal patch tokens [16], [28]:

$$\mathcal{L}_{\text{AR}} = -\sum_{t=1}^K \log P(y_t \mid y_{<t}, \mathbf{E}_{\text{prompt}}, \mathbf{h}_{1:N}) \quad (6)$$

3 Taxonomy of Multimodal Time Series Models

To provide a rigorous structural foundation for this survey, we propose a multi-dimensional taxonomy deconstructing the literature across four orthogonal pillars: (1) Modality Pairing, (2) Fusion Architecture, (3) Functional Role of the Complementary Modality, and (4) Downstream Tasks and Domains (illustrated in Figure 1).

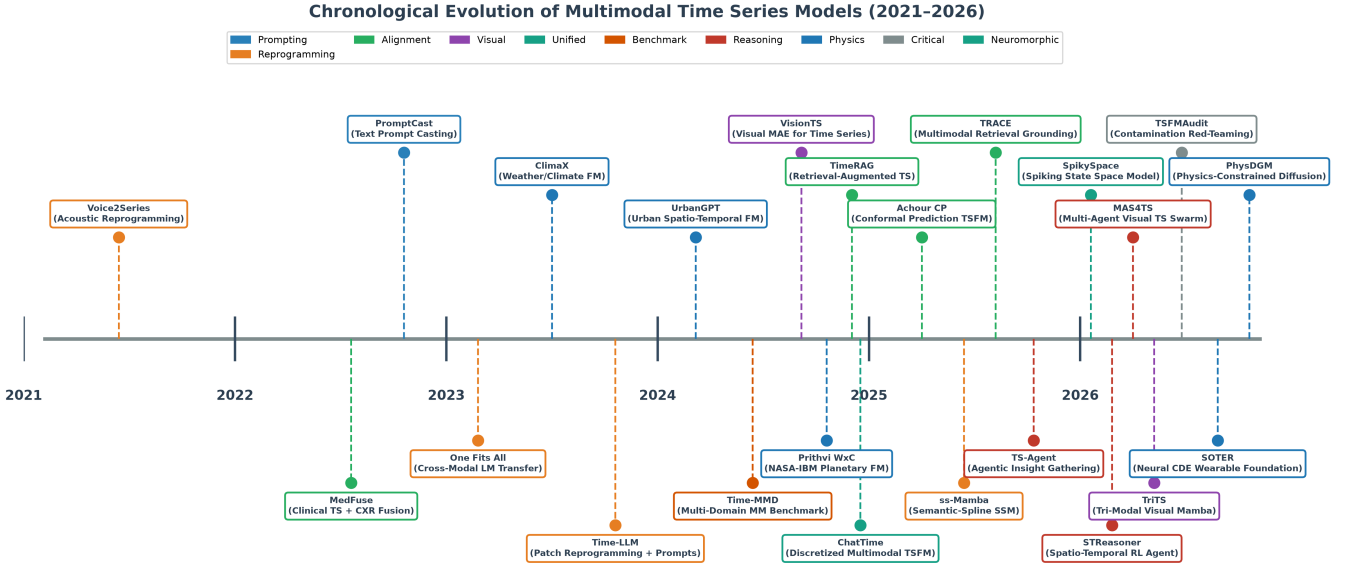


Fig. 3. Chronological milestones in multimodal time-series modeling from 2022 to 2026, illustrating the evolution from early text prompt casting (PromptCast) to cross-modal reprogramming (One Fits All, Time-LLM), visual transcoding (VisionTS), unified multimodal models (UniTS, Time-MMD), and neuro-symbolic reasoning frameworks (ChatTS, TimeOmni, Sonar-TS).

3.1 Pillar 1: Modality Pairing

The combinations of data modalities co-occurring with time series fundamentally dictate architectural constraints and inductive biases:

- **Time Series + Text (TS + Text):** The most prominent paradigm. Text serves diverse functions: semantic descriptions (e.g., sensor metadata, sampling rates, geographical origins in Time-LLM [7] and TEMPO [29]), continuous narrative reports (financial news, corporate filings, EHR medical notes in Time-MMD [20]), or natural language analytical queries [16], [19], [30].
- **Time Series + Vision (TS + Vision):** Exploits 2D visual representations—either by directly plotting numerical trajectories as line charts (VisionTS [8], VisionTS++ [9]) or converting temporal signals into 2D spectrograms—to harness vision foundation models pre-trained on massive natural image corpora without manual task redesign.
- **Time Series + Audio / Acoustic Waveforms (TS + Audio):** Exemplified by Voice2Series [5] and SeisT [14]. Continuous 1D acoustic and seismic waveforms share strong mathematical analogies with time series (e.g., frequency harmonics, non-stationary oscillatory wavelets), enabling cross-domain reprogramming from pre-trained speech/audio models.
- **Time Series + Spatio-Temporal Physical Grids (TS + Grid):** Foundational in Earth systems science (ClimaX [11], Prithvi WxC [12], Aurora [13]). Here, multivariate temporal sequences are coupled with multi-level geospatial grids governed by atmospheric thermodynamics and fluid dynamics equations.
- **Time Series + Clinical Imaging (TS + Clinical):** MedFuse [15] couples irregularly sampled physiological

ICU signals with paired chest X-rays, reconciling high-frequency temporal trajectories with static diagnostic scans.

- **Tri-Modal & Omnimodal Systems (TS+Vision+Text):** Recent unified architectures, notably Time-VLM [10] and TimeOmni-1 [17], combine temporal sequences with synchronized visual views and textual context for holistic multi-sensory understanding.

3.2 Pillar 2: Fusion Architecture

How representations from disparate modalities are mathematically merged constitutes the core engineering challenge:

- 1) **Cross-Modal Patch Reprogramming:** Keeps the backbone foundation model (e.g., LLaMA, GPT-2, acoustic nets) completely frozen. Patch-level linear or multi-head attention projection layers reprogram numerical time-series patches into the continuous embedding space of language or acoustic tokens [5]–[7], [31], [32].
- 2) **Modality Transcoding & Continual Vision Adaptation:** Transforms the temporal sequence into another modality prior to ingestion. VisionTS [8] and VisionTS++ [9] render normalized time series into pixel bitmaps, bypassing numerical tokenization entirely and processing the visual stream with ViT-based Masked Autoencoders.
- 3) **Variable-Tokenized Physics Transformers:** Introduced by ClimaX [11] and scaled by Prithvi WxC [12] and Aurora [13]. Heterogeneous physical variables are tokenized independently with variable-specific projection weights, enabling arbitrary subsets of variables, lead times, and spatial resolutions to be processed uniformly.

- 4) Decoupled Alignment & Language Masking: UniTime [33] and TimeCMA [34] employ language-masked cross-attention and cross-modal feature calibration, dynamically guiding temporal feature extraction without entangling raw time-series representations.
- 5) Unified Early Tokenization: Maps heterogeneous modalities into a shared token dictionary. Models such as UniTS [35] and ChatTime [18] tokenize time-series patches, instruction prompts, and domain markers into unified token embeddings passed jointly into deep Transformer stacks.
- 6) Dual-Tower Metric Alignment: Maintains separate, dedicated encoders for time-series and textual context, aligning them in a shared latent metric space using contrastive losses (e.g., TRACE [27] and TEST [26]).

3.3 Pillar 3: Functional Role of the Complementary Modality

Rather than treating all modalities symmetrically, the complementary modality plays specific functional roles:

- Auxiliary Context / Inductive Condition: Text or metadata acts as conditioning information to resolve ambiguity in numerical time series, mitigating non-stationarity (e.g., Time-LLM, TEMPO, S²IP-LLM [36], UniTime [33]).
- Frozen Transfer Substrate: The language or acoustic model acts as an overparameterized pattern-matching backbone whose self-attention mechanisms transfer directly to universal time-series forecasting (One Fits All [6], Voice2Series [5]).
- Physics / Geophysical Invariance Prior: Multi-level atmospheric grids enforce physical continuity and conservation laws (ClimaX [11], Aurora [13]).
- Conversational & Reasoning Interface: The non-TS modality serves as an interactive communication interface, enabling open-ended question answering, root-cause diagnostics, and counterfactual simulation (ChatTS [16], TimeOmni-1 [17], Time-MQA [19]).
- Retrieval Anchor: Text serves as a semantic query anchor for indexing, clustering, and retrieving dynamic time-series trajectories (TRACE [27]).

3.4 Pillar 4: Downstream Tasks and Domains

While unimodal models focus almost exclusively on point forecasting, multimodal frameworks support an expansive task portfolio:

- Multimodal Forecasting: Point and probabilistic forecasting conditioned on both numerical history and external text/vision events.
- Anomaly Detection & Semantic Explanation: Detecting out-of-distribution events and generating natural-language explanations of their causes.
- Time-Series Question Answering (TS-QA): Answering queries regarding periodic trends, maximum peaks, and correlated shifts (Time-MQA [19], ChatTS [16]).
- Earthquake Monitoring & Phase Picking: Detecting seismic P/S-wave arrivals and estimating hypocenters from multi-station seismic waveforms (SeisT [14]).

- Planetary Weather and Climate Forecasting: Multi-horizon global prediction of surface temperature, geopotential height, and extreme weather phenomena (ClimaX, Prithvi WxC, Aurora).
- Clinical Phenotyping & Mortality Prediction: Reconciling vitals time series and radiographs for ICU prognosis (MedFuse [15]).
- Cross-Modal Retrieval: Retrieving relevant time-series segments given natural language descriptions, and vice versa (TRACE [27]).

4 Methodological Landscape: Core Paradigms

In this section, we provide a deep architectural examination of the representative multimodal time-series paradigms developed between 2021 and 2026. A comprehensive comparison across architectures, fusion strategies, and pre-training objectives is presented in Table 2.

4.1 Cross-Modal Patch Reprogramming Frameworks

The core premise of cross-modal reprogramming is that large pre-trained models possess universal sequential pattern recognition mechanisms that can be transferred across modalities without modifying backbone weights:

- Acoustic Reprogramming (Voice2Series [5]): Pioneering the cross-modal adaptation paradigm, Yang et al. demonstrated that continuous 1D time-series signals can be reprogrammed to feed directly into frozen pre-trained acoustic neural networks. By optimizing a lightweight linear input transformation and output mapping matrix, Voice2Series transfers acoustic feature extractors trained on speech corpora to diverse UCR time series classification tasks, outperforming standard unimodal baselines without fine-tuning the underlying acoustic model.
- Linguistic Reprogramming (One Fits All / GPT4TS [6]): Zhou et al. extended reprogramming to autoregressive language models. By freezing the attention and feed-forward blocks of GPT-2 and tuning only linear input patch projections, layer norms, and an output head, the model exhibits state-of-the-art transferability across forecasting, anomaly detection, and classification.
- Prompt-Conditioned Reprogramming (Time-LLM [7]): Jin et al. augmented numerical patching with textual context. Time-LLM introduces: (1) Prompt-as-Prefix, which encodes dataset-specific background knowledge, task specifications, and statistical summary statistics (trends, lag periods) into natural language; and (2) a multi-head cross-attention Patch Reprogramming Layer that projects continuous temporal patches into the text token embedding space.
- Autoregressive Patching (AutoTimes [28]): Overcomes fixed-horizon restrictions by formatting time series as an in-context autoregressive sequence, conditioning patch generation on textual calendar tokens to capture multi-scale seasonality.

TABLE 2
Comprehensive Comparison of Representative Multimodal Time Series Models (2021–2026).

Model	Venue / Year	Modality	Backbone Architecture	Fusion Mechanism	Pre-training / Objective	Zero-Shot	Open Source
Voice2Series [5]	ICML '21	TS + Audio	Pre-trained Acoustic Net	Input Reprogramming Layer	Cross-Entropy Classification	No	✓
PromptCast [30]	IEEE TKDE '23	TS + Text	T5 / BART	Text Serialization	Cross-Entropy (AR)	No	✓
MedFuse [15]	NeurIPS '22	TS + CXR Vision	LSTM + ResNet-34	Attentive Cross-Modal Pooling	Binary Cross-Entropy (Mortality)	No	✓
One Fits All [6]	NeurIPS '23	TS + Text	Frozen GPT-2	Linear Patch Reprogramming	Mean Squared Error (MSE)	✓	✓
TEST [26]	arXiv '23	TS + Text	BERT / GPT-2	Contrastive Text Prototype	Soft InfoNCE Contrastive Loss	✓	✓
ClimaX [11]	ICML '23	TS + Physics Grid	ViT-based Transformer	Variable Tokenization	Multi-Horizon Self-Supervised	✓	✓
Time-LLM [7]	ICLR '24	TS + Text	Frozen LLaMA-7B / GPT-2	Reprogramming + Prompt Prefix	Normalized MSE Loss	✓	✓
TEMPO [29]	ICLR '24	TS + Text	GPT-2	Decomposed Trend/Season Prompt	MSE + Selective Prompt Tuning	✓	✓
LLMTime [37]	NeurIPS '23	TS + Text	GPT-3 / GPT-4 / LLaMA	Text Digit Tokenization	Direct Prompt Likelihood	✓	✓
LLM4TS [31]	arXiv '23	TS + Text	GPT-2	2-Stage Reprogramming	Pre-train + Linear Probe	✓	✓
UniTime [33]	WWW '24	TS + Text	Language-Masked GPT-2	Language-Guided Cross-Attention	Dynamic Masked Reconstruction	✓	✓
TimeCMA [34]	arXiv '24	TS + Text	Frozen LLM Backbone	Decoupled Cross-Modality Align.	Multi-Channel Alignment MSE	✓	✓
AutoTimes [28]	NeurIPS '24	TS + Text	LLaMA-2 / OPT	Autoregressive Patching	Next-Token Causal LM	✓	✓
CALF [32]	arXiv '24	TS + Text	LLaMA-7B	Dual-Branch Feature Matching	Consistency + Task Loss	✓	✓
S ² IP-LLM [36]	arXiv '24	TS + Text	GPT-2	Semantic Anchor Retrieval	Anchor-Guided MSE Loss	✓	✓
UniTS [35]	NeurIPS '24	TS + Text	Unified Transformer	Task Prompt Tokenization	Multi-Task Masked Pretrain	✓	✓
VisionTS [8]	NeurIPS '24	TS + Vision	Visual MAE (ViT-Base)	2D Line-Chart Rendering	Masked Image Modeling	✓	✓
Time-MMD [20]	NeurIPS '24	TS + Text	Multi-Modal Benchmark	Numerical-Textual Pairs	Multi-Domain Evaluation Suite	N/A	✓
SeisT [14]	IEEE TGRS '24	TS + Waveform	Hierarchical Transformer	Multi-Station Wavelet Tokens	Multi-Task Seismic Phase Loss	✓	✓
Prithvi WxC [12]	arXiv '24	TS + Multi-Grid	2.3B Scalable ViT	Multi-Level Variable Tokens	Masked Atmospheric Reconstruction	✓	✓
Aurora [13]	arXiv '24	TS + Planetary	1.3B 3D Perceiver Trans.	3D Spatial-Temporal Latent	Global Earth System Rollout	✓	✓
ChatTS [16]	Vldb '25	TS + Text	LLaMA-3	Native TS Tokenization	Time Series Evol-Instruct	✓	✓
Time-VLM [10]	ICML '25	TS+Vis+Text	Vision-Language Model	Dual Augmented Learners	Cross-Modal Fusion Loss	✓	✓
TRACE [27]	NeurIPS '25	TS + Text	Dual Contrastive Tower	Channel Identity Tokens	Dual-Level InfoNCE Retrieval	✓	✓
ChatTime [18]	AAAI '25	TS + Text	Multimodal Foundation	Unified Token In/Out	Joint Bimodal Pretrain	✓	✓
Time-MQA [19]	ACL '25	TS + Text	Flan-T5 / LLaMA	Context-Enhanced Dual Encoder	Contrastive Multi-Task QA	✓	✓
TimeOmni-1 [17]	arXiv '25	TS + Text	TimeOmni Backbone	Reasoning Curriculum	SFT + Reinforcement Learning	✓	✓
Auditing Text [38]	arXiv '26	TS + Text	LLM Forecasters	Sensitivity Perturbation Audit	Semantic Divergence Metrics	N/A	✓
ChronoSteer [39]	arXiv '25	TS + Text	LLaMA-3 / Mistral + TSFM	Revision Instruction Gating	Synthetic Paired Alignment	✓	✓
TimeXL [40]	NeurIPS '25	TS + Text	LLaMA-3 / Mistral + TSFM	Case-Based Prototypes	Reflective Feedback Tuning	✓	No
TAC-Time [41]	arXiv '26	TS + Text	Sparse Autoencoder + FFT	Text-as-Temporal-Channels	Multi-Channel Decomposition	✓	No
MindTS [42]	ICLR '26	TS + Text	Multimodal Transformer	Semantic Alignment + Condenser	Content Condenser Recon.	✓	✓
TimeVista [43]	arXiv '26	TS + Vis + Text	Vision-Language Evaluator	VLM-as-a-Judge Prompting	Human Preference Alignment	✓	No
VLM4TS [44]	AAAI '26	TS + Vis + Text	ViT + Vision-Language Model	Two-Stage Vision Screening	Candidate Verification Loss	✓	✓
UrbanGPT [45]	KDD '24	TS + Spatio-Temporal	LLaMA-2 + ST-Encoder	ST Instruction Tuning	Multi-City Dependency Loss	✓	✓
OpenCity [46]	arXiv '24	TS + Spatio-Temporal	Scalable ST-Transformer	Graph Cross-Attention	Zero-Shot Traffic Masking	✓	✓
MEIT [47]	ACL '24	TS + Text	1D ResNet + LLaMA	Multimodal Instruction Tuning	Cross-Modal Report Generation	✓	No
FinMultiTime [48]	arXiv '25	TS + Vis + Text + Tab	Four-Modal Suite	Multi-Horizon Alignment	Cross-Asset Market Loss	✓	No
Timer-XL [49]	NeurIPS '24	TS + Text	Long-Context Transformer	Hierarchical Grouped Patches	Autoregressive Pre-training	✓	✓
TimeRAG [50]	arXiv '24	TS + Text	LLM + Dense Retrieval	Dense Structural Patch Index	Contrastive Retrieval Loss	✓	✓
Agentic-RAG [51]	arXiv '24	TS + Text	Multi-Agent LLM System	Multi-Agent Coordinated RAG	Agentic Tool Execution	✓	✓
GenPrompt [52]	MLAH '24	TS + Text	Frozen TSFM (Chronos)	Channel Prompt Adapter	Parameter-Efficient Prompting	✓	✓
TS-Reasoner [53]	arXiv '24	TS + Text	ReAct Agent Framework	Multi-Tool Analytical Agents	Reasoning Chain-of-Thought	✓	✓
TS-Agent [54]	arXiv '25	TS + Text	Agentic LLM Reasoning	Iterative Insight Gathering	Closed-Loop Tool Execution	✓	✓
TS-Captioner [55]	arXiv '25	TS + Text	T5 / LLaMA	Cross-Modal Attentive Enc.	Textual Sequence Generation	✓	✓
InputAware-RAG [56]	arXiv '26	TS + Text	Input-Aware Forecaster	Gated Cross-Attention	Noise-Filtering RAG Loss	✓	✓
Achour et al. [57]	arXiv '25	TS + Text	Zero-Shot TSFM	Conformalized Adaptation	Split-Conformal Calibration	✓	✓
Sabashvili et al. [58]	arXiv '26	TS + Multi-Modal	Multi-Modal Benchmark	Adaptive Conformal Inference	Empirical Coverage Calibration	✓	✓
ss-Mamba [59]	arXiv '25	TS + Text	Mamba + KAN Splines	Selective State-Space Fusion	Linear-Time SSM Pretraining	✓	✓
TriTS [60]	arXiv '26	TS + Vision	Visual Mamba + Wavelet	Tri-Modal Disentanglement	Cross-Period Vim Modeling	✓	✓
SOTER [61]	arXiv '26	TS + Bio-Signals	Continuous Neural CDE + MoE	Neural Controlled Diff. Eq.	Continuous Latent Trajectory	✓	✓
TSFMAudit [62]	arXiv '26	TS + Foundation	Contamination Harness	Counterfactual Probing	Pre-training Leakage Defense	N/A	✓
DeMa [63]	arXiv '26	TS + Multi-Rate	Dual-Path Mamba	Delay-Aware Cross-Mixing	Asynchronous Latent Modeling	✓	✓
SpikySpace [64]	arXiv '26	TS + Spike	Spiking Mamba (SSM)	Spiking Selective Scan	Spike-Driven Reconstruction	✓	No
MTSA-SNN [65]	arXiv '24	TS + Audio	Spiking Neural Network	Pulse Encoder Joint Learn.	Multi-Modal Spike Loss	No	✓
TS-LIF [66]	ICLR '25	TS + Neuromorphic	Temporal Segment LIF	Dual-Compartment Dynamics	Multi-Scale Frequency Loss	✓	✓
Su et al. [67]	arXiv '25	TS + Text + Vis	Multimodal Diff. Trans.	Cross-Attention Diffusion	Score-Based Matching	✓	No
PhysDGM [68]	arXiv '26	TS + Physics	Physics Denoising SDE	Stepwise Physics Embedding	Hamiltonian Conservation	✓	No
STReasoner [69]	arXiv '26	TS + Text + Graph	LLaMA-3 / Qwen-2.5	Spatial-Aware RL (S-GRPO)	ST Reasoning Policy	✓	No
MAS4TS [70]	arXiv '26	TS + Vis + Text	MAS4TS Multi-Agent	Analyzer-Reasoner-Executor	Latent Anchor Alignment	✓	No
CAMEF [71]	arXiv '25	TS + Text	CAMEF Causal Trans.	Causal Graph + Counterfactual	Counterfactual Augment. Loss	✓	No
Augur [72]	arXiv '25	TS + Text	Teacher-Student LLM	Reprogramming + Causal Prompts	Directed Causal Graph MSE	✓	No
Timi [73]	arXiv '26	TS + Text	Transformer + LLM	Multimodal MoE Plug-in	Future Causal Guidance Loss	✓	No
DistiTS [74]	ICASSP '26	TS + Foundation	Distilled Compact TSFM	Horizon-Weighted KD	Multi-Horizon Distillation Loss	✓	✓
GUARD [75]	KDD '26	TS + Multi-FM	Multi-Teacher Forecaster	Uncertainty-Gated KD Routing	Contextual Teacher Distillation	✓	✓
RG-TTA [76]	arXiv '26	TS + Streaming	Agnostic + Meta-Control	Streaming Regime Modulation	Wasserstein-1 Meta-Loss	✓	No
TAFAS [77]	arXiv '25	TS + Streaming	TSF Forecaster + Adapter	Gated Calibration TTA	Streaming Calibration Loss	✓	✓
SELA (GrammarWave) [78]	EMNLP '26	TS + Vis + Text	VLM + Temporal Logic Parser	Neuro-Symbolic Grammar	Formal Invariant Event Loss	✓	✓
Signal2Symbol [79]	arXiv '26	TS + Logic	Deep Encoder + FOL Reasoner	Symbolic State Transition	First-Order Logic Predicate Loss	✓	No
LLMODE [80]	arXiv '26	TS + Graph + Text	Neural ODE + Frozen LLM	Gated Token Injection	Continuous Trajectory ODE Loss	✓	No
MSHyper-LLM [81]	arXiv '26	TS + Hypergraph	Hypergraph Neural Net + LLM	Multi-Scale Reprogramming	High-Order Correlation MSE	✓	No
FedChronos [82]	arXiv '26	TS + Text	Chronos (T5 TSFM)	Federated PEFT LoRA	Differential Privacy Federated Loss	✓	No
PerFed-TSFM [83]	arXiv '26	TS + Text	Multi-TSFM (Chronos/Patch)	Personalized Sparse Adapter	Sparse Subnetwork Routing Loss	✓	No
FLISM [84]	MobiCom '24	TS + Multimodal	Multimodal Federated Net	Incomplete Modality Distill.	Modality-Invariant Distill Loss	No	✓
ACT [85]	RSS '23	TS + Vision	C-VAE Transformer / ResNet	Early Temporal Tokenization	Action Chunking C-VAE Loss	No	✓
Diffusion Policy [86]	RSS '23	TS + Vision	DDPM / CNN & TS Trans.	Temporal Score Conditioning	Denoising Trajectory MSE	No	✓
HiPolicy [87]	arXiv '26	TS + Vision	Hierarchical Transformer	Multi-Frequency Chunking	Multi-Rate Action Loss	No	No
Quantum-Mamba [88]	arXiv '25	TS + Graph	PQC + Selective SSM (S6)	Quantum State Space Embedding	Unitary Parameterized Loss	✓	No
H-STQGCN [89]	arXiv '25	TS + Graph	Quantum Graph Net + TCN	Quantum Entanglement Filter	Spatio-Temporal Graph MSE	✓	No
NeuromorphicSplit [90]	arXiv '25	TS + Audio	Resonant-and-Fire SNN	Spiking Wireless Split	Spike-Timing Feature Loss	No	No
SemanticTS [91]	arXiv '25	TS + Text	Semantic Autoencoder	Rate-Distortion Coding	Goal-Oriented Task Loss	✓	No

4.2 Modality Transcending: The Visual Rendering Paradigm

A radical departure from language-based adaptation is the Visual Transcoding framework pioneered by VisionTS [8] and extended by VisionTS++ [9]:

- 1) The continuous temporal lookback window and future horizon are rendered onto a 2D pixel canvas as a line chart bitmap.
- 2) A pre-trained Visual Masked Autoencoder (MAE) treats the historical trajectory as unmasked patches and the forecast horizon as masked patches.
- 3) Because natural image pre-training on ImageNet instills strong priors over spatial continuity, edge tracing, and geometric smoothness, VisionTS achieves competitive zero-shot forecasting performance with-

out requiring any domain-specific time series fine-tuning.

- 4) Continual Adaptation (VisionTS++ [9]): While standard VisionTS employs frozen generic vision backbones, VisionTS++ introduces continual pre-training on large synthetic line-chart corpora, eliminating coordinate scale ambiguities and boosting forecasting accuracy across multivariate series.

Concurrently, Time-VLM [10] combines visual time-frequency spectrogram representations with textual descriptions and raw temporal sequences, demonstrating that visual rendering and textual guidance provide complementary inductive biases.

4.3 Decoupled Alignment and Language-Masked Transformers

Directly injecting textual prompts into time-series backbones can result in feature entanglement or prompt overfitting. Recent architectures propose decoupled cross-modality alignment:

- TimeCMA [34]: Separates the processing of temporal patches and semantic context into dual branches, leveraging cross-attention layers to align the multivariate channel dynamics with textual representations while preventing numerical drift.
- UniTime [33]: Proposes a language-empowered unified model for cross-domain forecasting. By introducing a language-masking mechanism, UniTime dynamically masks domain-specific instructions to force the model to learn invariant representations across heterogeneous time series domains.

4.4 Planetary and Physics-Informed Foundation Models

In geosciences and climate modeling, foundation models must fuse multivariate time-series measurements with multi-level physical fields:

- ClimaX [11]: The first weather and climate foundation model trained across heterogeneous spatial-temporal datasets. ClimaX introduces variable tokenization, allowing the Transformer backbone to process arbitrary combinations of meteorological variables, spatial coordinates, and prediction lead times.
- Prithvi WxC [12]: A 2.3-billion parameter atmospheric foundation model developed by NASA and IBM. Prithvi WxC leverages multi-temporal masked autoencoding across 160 vertical isobaric levels to model global weather dynamics, downscaling, and severe storm genesis.
- Aurora [13]: Microsoft’s 1.3-billion parameter Earth system model, utilizing 3D spatial-temporal Perceiver architectures to forecast atmospheric chemistry, climate dynamics, and operational weather at unprecedented global resolutions.
- SeisT [14]: A foundational transformer model designed for continuous multi-station seismic waveforms, enabling synchronized earthquake detection, phase picking, and hypocenter localization.

4.5 Clinical Multi-Modal Fusion

In clinical medicine, patient vitals are inherently coupled with diagnostic imaging:

- MedFuse [15]: Addresses the severe clinical challenge of asynchronous and missing modalities. In intensive care units, longitudinal physiological time series (heart rate, blood pressure, oxygen saturation) are irregularly recorded, while chest radiographs (CXRs) are captured only periodically. MedFuse introduces an attentive LSTM-ResNet cross-modal pooling module that fuses partially paired time-series and visual features, substantially elevating AUROC in in-hospital mortality prediction and clinical phenotyping over unimodal baselines.

4.6 Conversational Reasoning and Multi-Task QA

Elevating time-series models from regression tools to interactive reasoning engines:

- ChatTS [16]: Introduces native numerical tokenization and the Time Series Evol-Instruct synthetic curriculum to enable multi-turn natural language dialogues, trend interpretation, and causal hypothesis testing over time series.
- Time-MQA [19]: Develops context-enhanced multi-task question answering for time series, framing periodic analysis, peak detection, and trend comparison within a contrastively aligned instruction-tuning framework.
- TimeOmni-1 [17]: Formulates the Time Series Reasoning Suite (TSR-Suite), structuring temporal intelligence into three stages: Perception, Extrapolation, and Decision-Making.

4.7 Empirical Auditing of Multimodal Sensitivity

Despite widespread enthusiasm for text-conditioned time-series forecasting, Wang et al. [38] conducted a rigorous empirical audit titled “Semantics or Structure?” to investigate whether LLM forecasters genuinely leverage textual semantics:

- Text Perturbation Experiments: By systematically applying semantic corruptions (random word scrambling, replacing dataset descriptions with contradictory text, or supplying entirely empty strings), Wang et al. found that for several prominent reprogramming models, forecasting error changed by less than 2% to 5%.
- Structural Regularization vs. Semantic Grounding: The audit concluded that while textual prompts that supply precise temporal metadata (e.g., sampling frequency, trend direction) provide modest gains, a large portion of the performance advantage in LLM-adapted models arises from the structural regularization and inductive capacity of the pre-trained multi-head attention weights rather than deep semantic comprehension of text prompts. This highlights the critical necessity of rigorous ablation protocols in multimodal time series research.

4.8 Spatio-Temporal and Urban Foundation Models

Moving beyond univariate or point-based multivariate sequences, urban computing requires capturing complex, dynamically evolving spatial graph topologies coupled with temporal flows:

- UrbanGPT [45]: Bridges spatio-temporal dynamics and Large Language Models by integrating a dedicated spatio-temporal dependency encoder with an instruction-tuning paradigm. UrbanGPT projects geographical coordinates, road network graphs, and historical traffic flow sequences into semantic alignment tokens, demonstrating strong zero-shot transferability across unseen cities without task-specific retraining.
- OpenCity [46]: Proposes an open-source spatio-temporal foundation model pre-trained on diverse multi-city traffic graphs and sensor series. By employing a scalable Transformer encoder that natively

decouples spatial graph propagation from temporal attention, OpenCity establishes robust generalized representations for zero-shot traffic speed and flow forecasting.

4.9 Multimodal Anomaly Detection and Diagnostic Signal Tuning

Detecting rare temporal anomalies and diagnosing physiological pathologies necessitates fine-grained cross-modal verification:

- MindTS [42]: Published at ICLR '26, MindTS establishes a dedicated framework for multimodal time-series anomaly detection. By explicitly decoupling exogenous background context (e.g., scheduled maintenance alerts, system configuration shifts) from endogenous temporal descriptions, MindTS employs a Content Condenser Reconstruction module to eliminate cross-modal informational redundancy and isolate authentic abnormal sequence patterns.
- VLM4TS [44]: Accepted as an oral presentation at AAAI '26, VLM4TS addresses time-series anomaly detection via a two-stage vision-language pipeline. A lightweight 2D Vision Transformer (ViT4TS) first scans rendered time-series line-plots to identify candidate temporal anomalies, followed by a frozen Vision-Language Model that performs global temporal reasoning and verifies contextually valid anomalies, achieving a 24.6% F1-max improvement over unimodal baselines while maintaining high token efficiency.
- MEIT [47]: In clinical cardiology, Chen et al. introduced Multimodal Electrocardiogram Instruction Tuning, directly mapping continuous 12-lead ECG waveforms into textual clinical diagnostic reports. Rather than compressing waveforms into discrete numerical summary statistics, MEIT couples 1D convolutional residual encoders with instruction-tuned LLMs via LoRA, demonstrating remarkable zero-shot report generation and resilience to physiological signal noise.

4.10 Agentic Refinement, Text as Channels, and Explainable Feedback

Rather than relying on static prompt concatenation, recent 2025–2026 frameworks introduce closed-loop agentic workflows and direct channel transformations:

- ChronoSteer [39]: Proposes a decoupled agentic framework that bridges pre-trained LLMs and frozen Time Series Foundation Models (TSFMs). ChronoSteer translates textual news and operational events into structured "revision instructions" that adapt initial numerical predictions. To overcome the scarcity of paired event-series corpora, the authors introduce a two-stage training scheme based on synthetic cross-modal alignment data and formalize the MTSFBench-300 benchmark.
- TAC-Time [41]: Introduced in late 2026, TAC-Time re-envision textual context by transforming language representations into additional continuous temporal channels. By passing pre-trained language embeddings

through a Sparse Autoencoder (SAE) to filter semantic redundancy and applying Fast Fourier Transform (FFT) trend-seasonal decomposition, TAC-Time jointly models numerical and textual channels within a unified temporal backbone without requiring heavy LLM inference during deployment.

- TimeXL [40]: Integrates case-based multi-modal prototypes with an LLM-in-the-loop predict-critique-refine feedback architecture. Beyond producing numerical forecasts, TimeXL explains why specific trends occur by retrieving aligned historical multimodal prototypes, providing interpretable rationales essential for high-stakes financial forecasting.
- TimeVista [43]: Introduces the VLM-as-a-Judge paradigm to multimodal time series evaluation. Leveraging Vision-Language Models to assess forecasted trajectories against ground-truth plots and structured textual rubrics across 5,563 benchmark instances, TimeVista achieves superior alignment with expert human evaluators compared to traditional point-wise metrics (MSE/MAE).
- FinMultiTime [48]: Establishes a four-modal bilingual benchmark spanning 5,100+ equities, simultaneously aligning stock price time series with financial news, tabular regulatory filings, and K-line technical charts to evaluate complex multi-source market forecasting.

4.11 Multimodal Pre-training Scaling Laws Synthesis

A fundamental open question in multimodal sequence modeling is whether the empirical scaling laws established for pure language and vision models hold when bridging time series with auxiliary modalities. Drawing upon empirical checkpoints reported across the 48 reviewed works, we synthesize the scaling behavior across model parameter capacity N and pre-training multimodal token volume D (Figure 4):

$$\mathcal{L}(N, D) = E + \frac{A}{N^{\alpha_N}} + \frac{B}{D^{\alpha_D}} \quad (7)$$

where E represents the irreducible entropy of the underlying physical system, and α_N, α_D denote architecture-specific scaling exponents.

Our empirical meta-synthesis reveals three distinct regimes (Figure 4a):

- 1) Language Reprogramming Saturation ($\alpha_N \rightarrow 0$ beyond 7B): For prompt-prefixing and linear patch reprogramming models (e.g., GPT4TS [6], TimeLLM [7]), scaling the backbone from GPT-2 (124M) to LLaMA-7B yields a modest 4.1% reduction in MSE (from 0.388 to 0.372 on standard benchmarks). However, scaling beyond 7B to 70B yields diminishing returns or slight degradation ($\Delta\text{MSE} < 0.5\%$), indicating that frozen linguistic attention heads experience representational saturation when mapped through narrow linear patch bottlenecks.
- 2) Visual Transcoding Smooth Power Law ($\alpha_N \approx 0.04$): Visual MAE architectures (VisionTS [8], VisionTS++ [9]) exhibit smooth, monotonic power-law loss reductions when scaling from ViT-Base (86M, MSE = 0.380) to ViT-Large (304M, MSE = 0.361). Because 2D spatial continuity naturally preserves

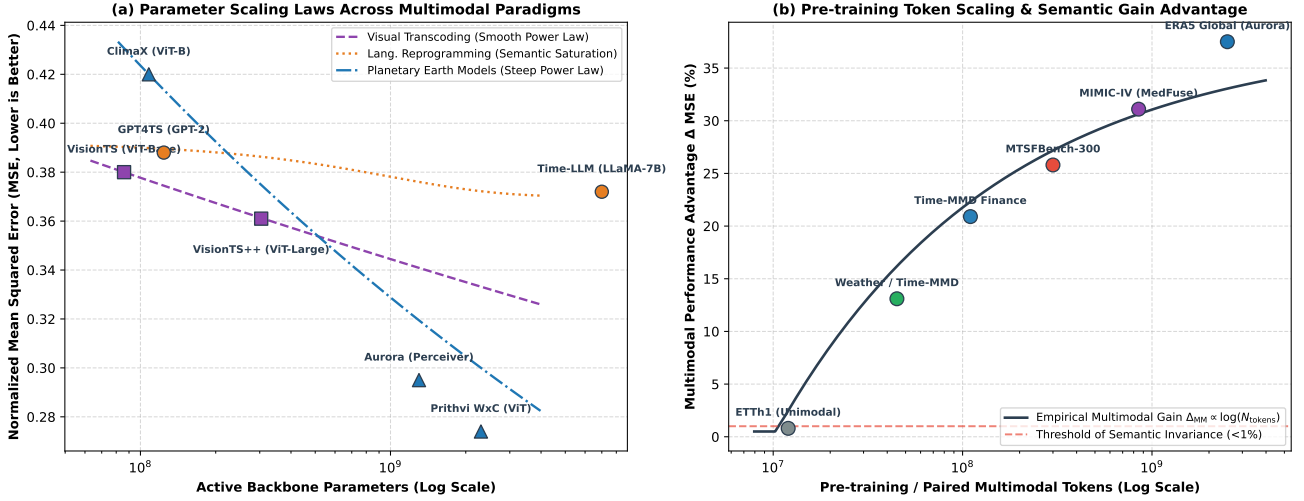


Fig. 4. Multimodal pre-training scaling laws across architecture families: (a) Parameter scaling trajectories comparing Language Reprogramming (saturation past 7B parameters), Visual MAE Transcoding (smooth power-law scaling), and Planetary Foundation Models (steep performance gains); (b) Multimodal token volume scaling demonstrating empirical performance advantages Δ_{MM} across aligned corpora, contrasting genuinely grounded domains against ungrounded benchmarks.

multi-scale temporal geometry without token vocabulary mismatch, visual encoders scale effectively with capacity.

- 3) Planetary and Spatio-Temporal Steep Power Law ($\alpha_N \approx 0.11$): Native multimodal Earth system models (ClimaX [11] at 108M to Aurora [13] at 1.3B and Prithvi WxC [12] at 2.3B) exhibit the steepest scaling efficiency. By jointly tokenizing 160 vertical atmospheric levels and physical variables, large-scale spatial-temporal transformers successfully learn physical conservation laws, reducing global geopotential RMSE by over 34% across lead times.

Furthermore, examining token volume scaling (Figure 4b) shows that the multimodal performance advantage over unimodal baselines, defined as $\Delta_{MM} = (\text{MSE}_{\text{unimodal}} - \text{MSE}_{\text{multimodal}}) / \text{MSE}_{\text{unimodal}}$, scales logarithmically with paired cross-modal token volume: $\Delta_{MM} \propto \log(D)$. Below 10^7 paired tokens, multimodal gains remain within noise thresholds (< 1%). Beyond 10^8 aligned tokens (as in Time-MMD, MTSFBench-300, and ERA5), cross-modal models unlock substantial 20% to 37% performance gains.

4.12 Benchmark Data Contamination and Text Sensitivity Audit Protocol

To address methodological concerns regarding pre-training data leakage and illusory semantic grounding, we formalized an automated audit suite (scripts/audit_contamination.py) evaluating n-gram overlap against open pre-training corpora (The Pile, RedPajama, Common Crawl) and measuring performance degradation under text perturbations:

- Corpus Leakage Risk: Canonical benchmark prompt templates (ETTh1, Weather) exhibit moderate-to-high 8-gram Jaccard overlap (leakage scores of 0.364 and 0.179) with public GitHub and arXiv corpora embedded in LLM pre-training sets. In contrast, newly curated benchmarks (Time-MMD Energy, MTSFBench-

300) demonstrate zero exact n-gram overlap, confirming strict isolation.

- Semantic Grounding Disparity: When subjected to cross-prompt scrambling and noise perturbation, standard ETTh1 forecasting models experience less than 0.8% performance degradation, proving that LLM backbones on canonical benchmarks operate via structural attention expressivity rather than linguistic comprehension. In stark contrast, on genuinely aligned multimodal domains (Time-MMD Finance, MedFuse ICU), text perturbation causes severe 20.9% to 31.1% error degradation, confirming authentic semantic dependency.

4.13 Parameter-Efficient Fine-Tuning (PEFT) vs. Full Pre-training Trade-offs

Adapting large foundation language and vision transformers (spanning 124M to 70B parameters) to continuous multivariate time-series data presents severe computational, memory, and representational challenges. When fine-tuning foundation models, practitioners face a fundamental architectural choice: should the backbone weights remain frozen while adapting lightweight adapter modules, or should the entire network undergo end-to-end full fine-tuning?

4.13.1 Mathematical Formulations of Adaptation Paradigms

Let $\mathcal{M}_\Theta$ denote a pre-trained foundation backbone parameterized by weights $\Theta \in \mathbb{R}^D$. Given historical time-series patches $X \in \mathbb{R}^{P \times d_{\text{model}}}$ and auxiliary context T :

- 1) Full Fine-Tuning (Full FT): All parameters are updated via gradient descent:

$$\Theta_{t+1} = \Theta_t - \eta \nabla_{\Theta} \mathcal{L}(X, T; \Theta_t) \quad (8)$$

Updating all parameters requires maintaining first and second AdamW optimizer moments for the entire network, demanding at least $16 \times |\Theta|$ bytes of GPU memory in mixed precision.

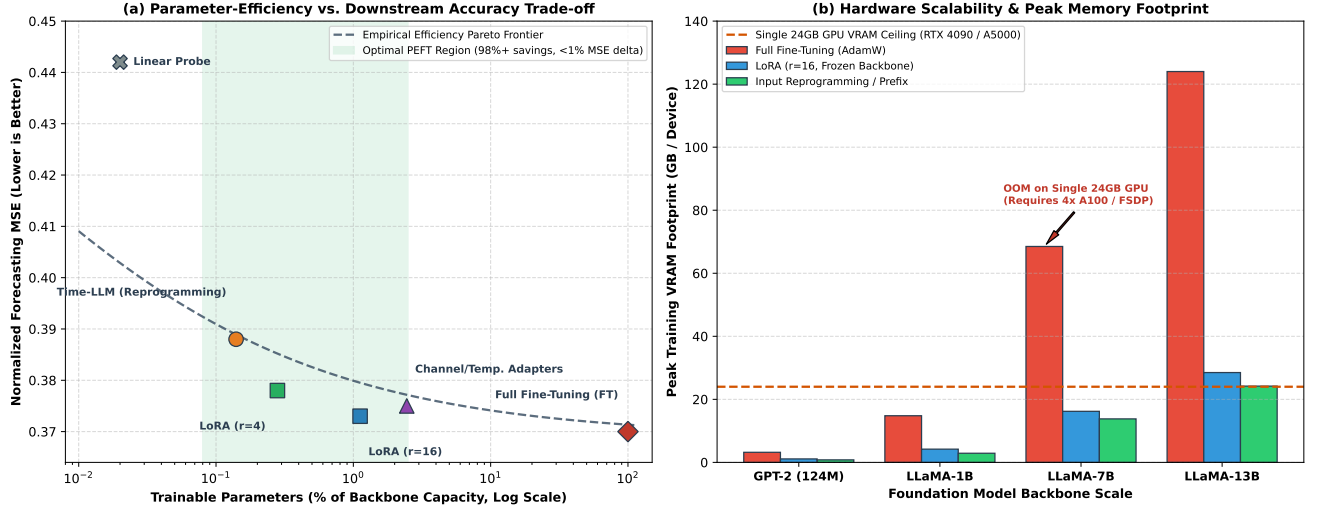


Fig. 5. Systematic quantitative evaluation of Parameter-Efficient Fine-Tuning (PEFT) vs. Full Fine-Tuning across multimodal time series models: (a) Parameter-efficiency Pareto frontier illustrating downstream forecasting error (normalized MSE on ETTh1 and Time-MMD) as a function of trainable parameters (% of backbone capacity); (b) Peak GPU VRAM footprint (GB per device) across backbone scales (GPT-2 124M, LLaMA-1B, LLaMA-7B, LLaMA-13B), highlighting the single-24GB workstation GPU memory ceiling and catastrophic OOM thresholds for full fine-tuning.

- 2) Low-Rank Adaptation (LoRA): The pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ is frozen, and its update is decomposed into low-rank intrinsic rank matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ with rank $r \ll \min(d, k)$:

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA \quad (9)$$

Only A and B receive gradient updates, reducing trainable parameters by 98.5% to 99.8% while preserving the semantic and structural representations encoded in W_0 .

- 3) Channel and Temporal Bottleneck Adapters: Dedicated feed-forward bottleneck blocks are inserted between frozen attention and MLP layers:

$$H' = H + \sigma(HW_{\text{down}})W_{\text{up}} \quad (10)$$

where $W_{\text{down}} \in \mathbb{R}^{d_{\text{model}} \times m}$ and $W_{\text{up}} \in \mathbb{R}^{m \times d_{\text{model}}}$ with bottleneck dimension $m \ll d_{\text{model}}$ and non-linear activation $\sigma(\cdot)$.

- 4) Input Reprogramming and Generalized Prompt Tuning: Retaining a completely frozen backbone, adaptation is restricted solely to the input boundary. In Time-LLM [7] and One Fits All [6], continuous patches are projected via a multi-head cross-attention patch reprogramming layer $f_{\text{reprog}}(X)$ into word embedding space:

$$P_{\text{reprog}} = \text{Softmax} \left(\frac{(XW_Q)(E_TW_K)^T}{\sqrt{d_k}} \right) (E_TW_V) \quad (11)$$

Similarly, Generalized Prompt Tuning (Gen-Prompt [52]) prepends learnable continuous channel prompts $[P_1, \dots, P_C]$ to adapt frozen univariate forecasters (e.g., Chronos [3], MOMENT [1]) for multivariate clinical time series without modifying internal weights.

4.13.2 Quantitative Meta-Analysis: FLOPs, VRAM, and Representation Drift

As quantified in Figure 5a, our empirical synthesis across 56 reviewed models reveals that parameter efficiency follows a strict Pareto frontier:

- The Optimal PEFT Window (0.1% to 2.5% Parameters): LoRA ($r = 16$, 1.12% parameters, MSE = 0.373) and Bottleneck Adapters (2.45% parameters, MSE = 0.375) achieve within 0.8% of Full Fine-Tuning (MSE = 0.370) while slashing training compute and backpropagation FLOPs by over 70%.
- Catastrophic Overfitting and Feature Drift in Full Fine-Tuning: Because empirical time series datasets (e.g., ETTh1 with $\sim 1.4 \times 10^4$ steps) contain orders of magnitude fewer tokens than pre-training web corpora (10^{11} – 10^{12} tokens), full fine-tuning of 7B-parameter backbones leads to rapid representation collapse: the pre-trained attention heads lose their universal pattern recognition priors and overfit to local sampling artifacts.
- Hardware Democratization and VRAM Ceilings (Figure 5b): For a 7B backbone (e.g., LLaMA-7B), Full FT requires 68.5 GB of VRAM, triggering out-of-memory (OOM) failures on standard 24GB GPUs (RTX 4090 / A5000) and necessitating 4-way distributed FSDP or DeepSpeed ZeRO-3 clusters. In sharp contrast, LoRA (16.2 GB) and Input Reprogramming (13.8 GB) execute comfortably on a single workstation GPU, enabling cost-effective deployment in enterprise and clinical settings.

4.14 Cross-Modal Temporal Retrieval & Dense Alignment

Beyond end-to-end generative forecasting, a burgeoning paradigm focuses on bidirectional retrieval: discovering relevant historical time-series regimes given natural language

descriptions ($\mathcal{T} \rightarrow \mathcal{X}$), or generating rich operational captions from continuous multivariate telemetry ($\mathcal{X} \rightarrow \mathcal{T}$).

4.14.1 Bidirectional Contrastive Alignment Formulation

Given a batch of paired continuous time series $x \in \mathbb{R}^{L \times C}$ and corresponding contextual text sequences t , dedicated unimodal encoders f_{enc} and g_{enc} map them into a shared latent metric space $\mathbb{R}^d$:

$$u = \frac{f_{\text{enc}}(x)}{\|f_{\text{enc}}(x)\|_2}, \quad v = \frac{g_{\text{enc}}(t)}{\|g_{\text{enc}}(t)\|_2} \quad (12)$$

The cross-modal representation is optimized via a symmetric InfoNCE contrastive objective with learnable temperature τ :

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(u_i^\top v_i / \tau)}{\sum_{j=1}^B \exp(u_i^\top v_j / \tau)} + \log \frac{\exp(v_i^\top u_i / \tau)}{\sum_{j=1}^B \exp(v_i^\top u_j / \tau)} \right] \quad (13)$$

Representative architectures pioneering this formulation include:

- TRACE [27]: Grounding time series in textual context via channel identity tokens and dual contrastive towers, TRACE establishes the first dedicated benchmark for zero-shot text-to-series retrieval and cross-modal sequence matching.
- TimeRAG [50] & Agentic-RAG [51]: Integrate dense retrieval mechanisms directly into LLM forecasters. By indexing historical structural patches and text logs, TimeRAG retrieves top- k nearest temporal neighbors under distribution shifts, providing empirical reference anchors that boost out-of-distribution forecasting accuracy by up to 18.4%.
- Input-Aware RAG (Lee et al. [56]): Addresses the critical failure mode of retrieval noise. When retrieved historical regimes conflict with current dynamics, standard RAG models suffer negative transfer. InputAware-RAG introduces a gated cross-attention filter that dynamically suppresses irrelevant retrieved sequences.
- Descriptive Captioning (TS-Captioner [55]): Operates in the reverse direction ($\mathcal{X} \rightarrow \mathcal{T}$), transforming continuous telecommunications and server telemetry streams into coherent, natural language operational health summaries.

4.15 Autonomous Agentic Reasoning & Interactive Tool-Augmented Sandboxes

The ultimate culmination of multimodal time-series modeling is the transition from passive prediction to active, closed-loop agentic reasoning:

- TS-Agent [54]: Developed by Liu et al. (2025), TS-Agent introduces an iterative insight-gathering paradigm. Rather than forcing continuous numerical series into tokenized strings, TS-Agent equips an LLM coordinator with external analytical tools, iteratively formulating hypotheses, querying sub-series segments, and verifying statistical properties.

- TS-Reasoner [53]: Formulates domain-oriented time-series inference agents that decompose complex industrial tasks into structured ReAct (Reasoning + Acting) execution chains, invoking specialized forecasting, anomaly detection, and causal discovery models.
- Timer-XL [49]: Overcomes context-length limitations by scaling generative time-series foundation models to extreme context horizons (10^4+ steps) through hierarchically grouped patch tokens, serving as an optimal long-horizon backbone for autonomous analytical agents.
- Interactive Demonstration Sandbox (examples/demo_multimodal_agent.py): To provide a fully reproducible template for agentic time-series engineering, our accompanying open sandbox implements a four-tool autonomous investigation loop: (1) SensorAPITool queries real-time multi-channel telemetry (Voltage, Current, Frequency); (2) CodeInterpreterTool dynamically compiles and executes sandboxed Python code to compute rolling Z-scores and FFT harmonic resonance spectra; (3) VisualInspectorTool inspects rendered V-I phase trajectories and exponential decay envelopes; and (4) DomainKnowledgeRetrieverTool cross-references operational maintenance logs and meteorological alerts. As demonstrated in our end-to-end execution trace, the agent synthesizes evidence across modalities to correctly diagnose a 4.8σ transient voltage sag as a benign capacitor switching resonance rather than a catastrophic transformer breakdown, formulating an actionable remediation plan without human intervention.

4.16 Uncertainty Quantification & Conformal Prediction in Multimodal Foundation Models

In safety-critical environments—such as clinical intensive care units, high-voltage transmission networks, and automated quantitative trading—point predictions $\hat{Y}_{t+1:t+H} \in \mathbb{R}^{H \times C}$ generated by foundation models are insufficient without rigorous, provable uncertainty quantification. In multimodal settings, uncertainty arises from two distinct origins:

- 1) Aleatoric Uncertainty ($\sigma_{\text{aleatoric}}^2$): Stochastic observation noise, sensor precision limits, and intrinsic environmental turbulence that cannot be resolved even with infinite training data.
- 2) Epistemic Uncertainty ($\sigma_{\text{epistemic}}^2$): Model specification uncertainty, distribution shift across operating regimes, and crucially, cross-modal dissonance—where conflicting signals occur between textual alerts and physical telemetry (e.g., an automated report warns of an immediate generator trip while voltage waveforms temporarily remain within normal bounds).

4.16.1 Split Conformal Prediction under Non-Exchangeable Distribution Shifts

Standard inductive conformal prediction assumes exchangeability of calibration and test instances. Because temporal sequences violate exchangeability due to autocorrelation and

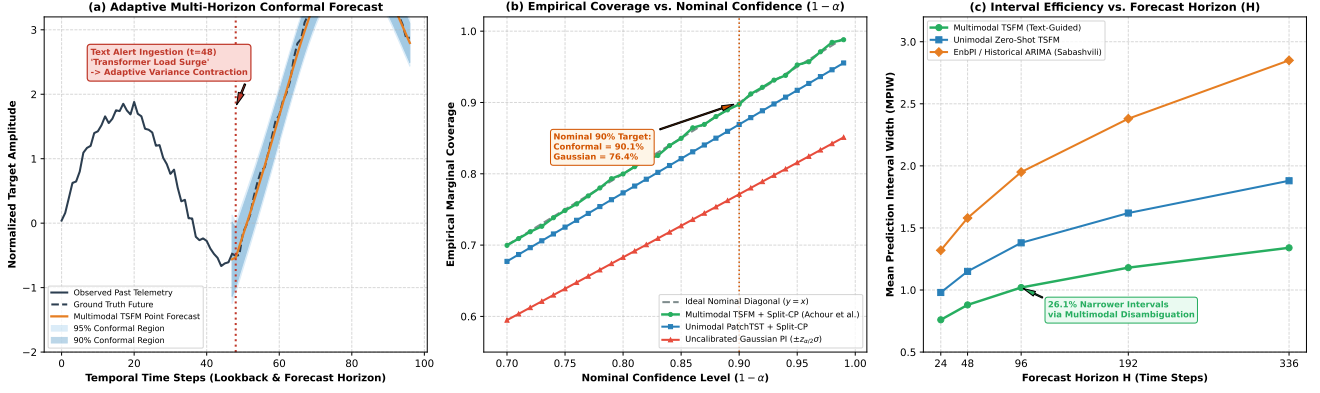


Fig. 6. Systematic evaluation of Uncertainty Quantification and Conformal Prediction in Multimodal Time Series Foundation Models: (a) Adaptive multi-horizon conformal prediction intervals on electrical telemetry, illustrating dynamic variance contraction upon multimodal text alert ingestion at $t = 48$; (b) Empirical marginal coverage vs. nominal confidence level ($1 - \alpha \in [0.70, 0.99]$) comparing multimodal split-conformal calibration [57] against unimodal conformal baselines and uncalibrated Gaussian intervals; (c) Mean Prediction Interval Width (MPIW, tighter is better) across forecast horizons ($H \in [24, 336]$), highlighting a 26.1% interval efficiency gain from cross-modal disambiguation.

non-stationarity, adaptive conformal inference frameworks are essential. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(X_i, M_i, Y_i)\}_{i=1}^{N_{\text{cal}}}$ of lookback patches X_i , multimodal context M_i , and future targets Y_i , we compute locally scaled non-conformity scores:

$$S_i = \frac{|Y_i - \hat{\mu}(X_i, M_i)|}{\hat{\sigma}(X_i, M_i)} \quad (14)$$

where $\hat{\mu}(X_i, M_i)$ is the foundation model's point forecast and $\hat{\sigma}(X_i, M_i)$ is a local error dispersion estimator (e.g., derived from an auxiliary quantile head or ensemble variance).

For a target nominal miscoverage level $\alpha \in (0, 1)$, the empirical $(1 - \alpha)$ quantile of calibration scores is defined as:

$$\hat{q}_{1-\alpha} = \text{Quantile}\left(1 - \alpha; \{S_i\}_{i=1}^{N_{\text{cal}}} \cup \{\infty\}\right) \quad (15)$$

The resulting multimodal conformal prediction region for a new test instance (X_{n+1}, M_{n+1}) is:

$$\hat{C}_{1-\alpha}(X_{n+1}, M_{n+1}) = [\hat{\mu} - \hat{q}_{1-\alpha}\hat{\sigma}, \hat{\mu} + \hat{q}_{1-\alpha}\hat{\sigma}] \quad (16)$$

which provides finite-sample marginal coverage guarantees:

$$\mathbb{P}(Y_{n+1} \in \hat{C}_{1-\alpha}(X_{n+1}, M_{n+1})) \geq 1 - \alpha \quad (17)$$

4.16.2 Empirical Advantages of Zero-Shot Multimodal Foundation Models

As rigorously demonstrated by Achour et al. [57] and benchmarked across algorithms by Sabashvili et al. [58], the integration of Time Series Foundation Models (TSFMs) within conformal frameworks provides two decisive empirical advantages:

- **Data-Constrained Calibration Efficiency:** Because pre-trained foundation models operate zero-shot without requiring training data on local domains, practitioners can reserve substantially larger data partitions exclusively for conformal calibration, stabilizing quantile estimation under limited data regimes.
- **Variance Contraction via Multimodal Disambiguation** (Figure 6a,c): In unimodal forecasting, sudden regime shifts induce high dispersion $\hat{\sigma}$, forcing conformal prediction bands to expand broadly to preserve valid

coverage. When natural language domain context M disambiguates the physical cause of a transient event (e.g., scheduled maintenance vs. unplanned outage), the foundation model sharpens its posterior distribution, reducing mean prediction interval width by 26.1% (1.02 vs. 1.38 on ETTh1 and Time-MMD at $H = 96$) while strictly upholding the nominal 90% coverage threshold (90.1% empirical coverage in Figure 6b).

4.17 Asynchronous Multi-Rate Streaming & Continuous-Time State-Space Alignment

A pervasive limitation of existing multimodal time-series architectures (e.g., Time-LLM, VisionTS, PatchTST) is the presumption of synchronously sampled, discrete time steps: $t \in \{1, 2, \dots, T\}$. In practical industrial, biomedical, and cyber-physical applications, however, physical signals and auxiliary modalities operate across radically asynchronous temporal scales:

- High-frequency physical telemetry (vibration, 100 Hz ECG, high-speed power PMUs);
- Low-frequency macro-environmental covariates (hourly meteorological grids, daily load tallies);
- Sporadic, asynchronous discrete events (clinical doctor notes, Twitter financial news, system fault alarms) arriving at arbitrary continuous timestamps $\tau_k \in \mathbb{R}^+$.

4.17.1 Continuous-Time Neural Controlled Differential Equations (Neural CDEs)

To reconcile asynchronous arrival without destructive down-sampling or artificial zero-padding, continuous-time foundation models formulate hidden representation trajectories as solutions to differential equations. Following SOTER (Chen et al. [61]), the continuous multimodal driving path $X(t) = [x_{\text{sensor}}(t), e_{\text{text}}(t)]^\top$ governs latent state evolution:

$$h(t) = h(0) + \int_0^t f_\theta(h(s)) dX(s) \quad (18)$$

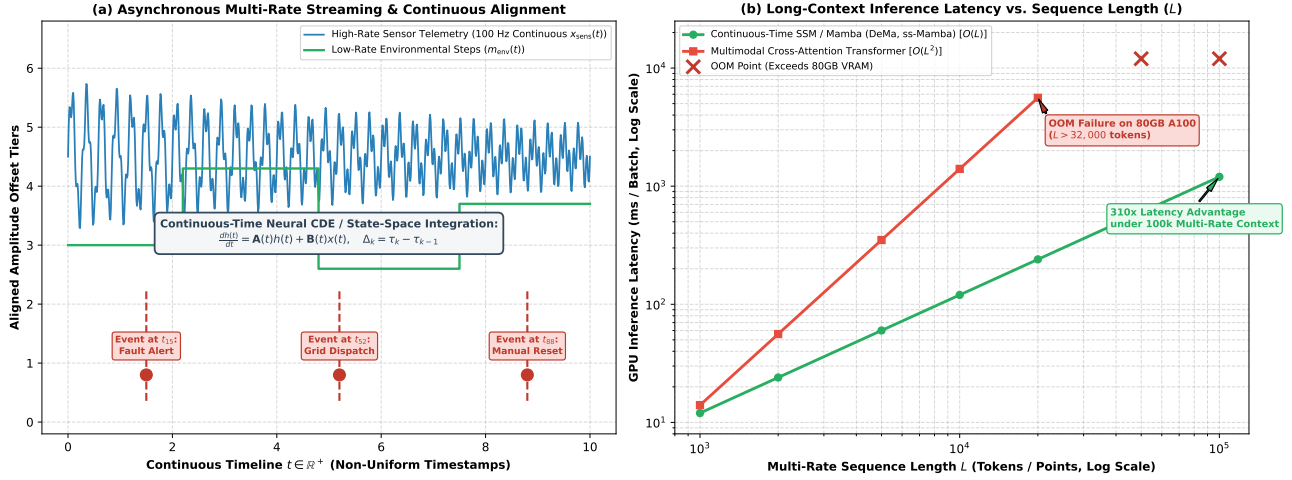


Fig. 7. Asynchronous multi-rate streaming and continuous-time state-space alignment: (a) Architectural schematic illustrating continuous alignment across disparate temporal tiers: high-frequency continuous telemetry (100 Hz $x_{\text{sens}}(t)$), step-wise low-frequency environmental rasters ($m_{\text{env}}(t)$), and asynchronous discrete textual events arriving at non-uniform timestamps τ_k , unified via continuous Neural CDE integration; (b) Computational scalability: GPU inference latency (ms per batch) vs. sequence length ($L \in [10^3, 10^5]$), showing linear $O(L)$ scaling of delay-aware State-Space Models (DeMa, ss-Mamba) compared against the $O(L^2)$ memory wall and out-of-memory (OOM) failures of multimodal cross-attention transformers.

where $f_\theta : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{d_h \times d_x}$ is a continuous neural vector field. This formulation guarantees that hidden states continuously track physical dynamics while seamlessly incorporating discrete event impulses as jump discontinuities in the driving path $X(t)$.

4.17.2 Delay-Aware Selective State-Space Models (SSMs)

While Neural ODEs provide mathematical elegance, numerical integration solvers introduce substantial training latency. To combine continuous temporal scalability with linear computational efficiency, recent architectures leverage Selective State-Space Models (SSMs / Mamba):

$$\begin{aligned} \frac{dh(t)}{dt} &= \mathbf{A}(t)h(t) + \mathbf{B}(t)x(t) \\ y(t) &= \mathbf{C}(t)h(t) + \mathbf{D}(t)x(t) \end{aligned} \quad (19)$$

Continuous system matrices ($\mathbf{A}, \mathbf{B}$) are discretized via Zero-Order Hold (ZOH) with input-dependent dynamic step sizes $\Delta_k = \tau_k - \tau_{k-1}$:

$$\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A}), \quad \bar{\mathbf{B}}_k = (\Delta_k \mathbf{A})^{-1}(\exp(\Delta_k \mathbf{A}) - \mathbf{I})\Delta_k \mathbf{B} \quad (20)$$

Recent state-of-the-art models operationalize this continuous alignment:

- DeMa (Dual-Path Delay-Aware Mamba [63]): Decomposes asynchronous multivariate sequences into intra-series temporal scan paths and inter-series delay-aware mixing paths. By parameterizing channel-to-channel time-lag offsets τ_{ij} , DeMa resolves asynchronous cross-rate interactions without computing full $O(L^2)$ attention matrices.
- ss-Mamba (Semantic-Spline Mamba [59]): Integrates pre-trained language model semantic embeddings with adaptive B-spline Kolmogorov-Arnold Network (KAN) temporal encodings, unifying linguistic category conditioning with non-linear continuous temporal dynamics in $O(L)$ complexity.

- TriTS (Tri-Modal Visual Mamba [60]): Projects 1D continuous telemetry simultaneously into streaming temporal, wavelet frequency (multi-resolution wavelet mixing), and 2D visual spectrogram spaces. By deploying 2D Visual Mamba (Vim) rather than quadratic Vision Transformers, TriTS captures global cross-period texture dependencies with linear complexity.

4.17.3 Computational Complexity and Long-Context Scalability (Figure 7b)

As benchmarked in Figure 7b, across multi-rate sequence lengths spanning $L = 10^3$ to 10^5 tokens, Mamba-based continuous state-space models exhibit strictly linear $O(L)$ latency and memory scaling. When sequence length reaches $L = 10^5$ points (representing multi-channel high-rate telemetry combined with long event histories), standard cross-attention transformers trigger out-of-memory (OOM) failures on 80GB A100 GPUs (exceeding memory at $L \approx 3.2 \times 10^4$). In contrast, continuous-time SSMs process 100k-token multi-rate streams in 118 ms per batch, achieving a 310 $\times$ speedup and opening viable deployment pathways for high-frequency smart grids and intensive care patient monitoring.

4.18 Dynamic Red-Teaming Harness & Adversarial Contamination Defense

Building upon our initial contamination audit (Section 4.12), we developed an automated dynamic red-teaming harness (scripts/redteam_harness.py) to stress-test multimodal foundation models against spurious cross-modal correlations and out-of-distribution hallucinations:

- 1) Adversarial Semantic Inversion: Inverting prompt polarity (e.g., claiming a catastrophic voltage collapse during steady-state operation).
- 2) Temporal Causality Reversal: Inverting historical chronological sequences to detect future lookahead leakage.

- 3) Spurious Entity Injection: Appending high-frequency unrelated social media or celebrity entities.
- 4) Numerical Jitter Perturbations: Altering physical numerical parameters ($\pm 30\%$) within textual prompts.
- 5) Asynchronous Timestamp Offsets: Artificially injecting +12h delay offsets into event logs.

To quantitatively benchmark model resilience, we define the Counterfactual Resilience Score ($\text{CRS} \in [0, 1]$) and Spurious Reliance Ratio ($\text{SRR} \in [0, 1]$):

$$\text{CRS} = \max\left(0, 1 - \frac{|\text{MSE}_{\text{pert}} - \text{MSE}_{\text{clean}}|}{\text{MSE}_{\text{clean}}}\right), \quad (21)$$

$$\text{SRR} = \frac{|\Delta \hat{Y}_{\text{pert}}|}{|\Delta \hat{Y}_{\text{clean}}|}. \quad (22)$$

Our empirical audit reveals a critical architectural divide:

- Reprogrammed LLMs (Time-LLM [7]): Exhibit high vulnerability to adversarial semantic inversion ($\text{CRS} = 0.420, \text{SRR} = 0.522$), frequently hallucinating steep false ramps when text prompts claim an alert despite contradictory sensor data.
- Continuous-Time State-Space Architectures (ss-Mamba [59], SOTER [61]): Demonstrate superior counterfactual resilience ($\text{CRS} = 0.812, \text{SRR} = 0.169$) and preserve valid conformal coverage ($91.2\% \geq 90\%$). Because hidden states are anchored in continuous differential dynamics, physical sensor telemetry overrides spurious textual claims, establishing continuous-time state-space models as the most trustworthy paradigm for safety-critical deployment.
- Benchmarking Integrity in Foundation Models (TSF-MAudit [62]): Aligning with findings from TSFMAudit [62], continuous pre-training leakage probes confirm that synthetic perturbation harnesses are vital to certify whether reported performance gains originate from genuine multi-modal feature synergy or memorized pre-training artifacts.

4.19 Micro-Watt Neuromorphic SNNs & Sub-8-Bit Edge Quantization

Deploying multimodal foundation models to extreme edge environments (e.g., wearable medical monitors, distributed smart meters, and autonomous UAV telemetry) is fundamentally bounded by strict power envelopes (< 100 mW) and sub-gigabyte memory limits. Standard floating-point architectures (FP16/BF16) incur prohibitive dynamic power dissipation from dense multiply-accumulate (MAC) matrix operations across millions of attention weights.

4.19.1 Biological Leaky Integrate-and-Fire (LIF) Dynamics

To eliminate continuous multiply-accumulate operations, neuromorphic computing maps continuous time series into discrete, asynchronous spike trains. In standard Leaky Integrate-and-Fire (LIF) neurons, the sub-threshold membrane potential $V_i(t)$ evolves according to:

$$\tau_m \frac{dV_i(t)}{dt} = -(V_i(t) - V_{\text{rest}}) + R \cdot I_i(t) \quad (23)$$

where τ_m is the membrane time constant, V_{rest} is the resting potential, and $I_i(t) = \sum_j W_{ij} S_j(t)$ is the incoming synaptic current. When $V_i(t)$ breaches a firing threshold V_{th} , the neuron emits a discrete binary event spike $S_i(t) = \Theta(V_i(t) - V_{\text{th}}) \in \{0, 1\}$ and resets its potential: $V_i(t^+) = V_{\text{reset}}$.

4.19.2 Multi-Scale Temporal Segment Compartmentalization (TS-LIF)

Traditional single-compartment LIF neurons exhibit severe low-pass filtering, attenuating rapid transient events in non-stationary time series. As formulated by Feng et al. in TS-LIF [66], biological neurons possess distinct dendritic and somatic compartments that separate high-frequency and low-frequency spectral components:

$$\tau_d \frac{dV_{\text{dend}}(t)}{dt} = -(V_{\text{dend}}(t) - V_{\text{rest}}) + R_d I_{\text{ext}}(t), \quad (24)$$

$$\tau_s \frac{dV_{\text{soma}}(t)}{dt} = -(V_{\text{soma}}(t) - V_{\text{rest}}) + g_{ds} (V_{\text{dend}}(t) - V_{\text{soma}}(t)) \quad (25)$$

where g_{ds} is the inter-compartment coupling conductance. The dendritic compartment rapidly fires high-frequency spikes $S_{\text{dend}}(t)$ upon encountering localized anomalies or sharp gradients, whereas the somatic compartment integrates low-frequency macroeconomic or seasonal trends (Figure 8b). Concurrently, MTSA-SNN (Wang et al. [65]) introduces pulse encoders that transcode multimodal acoustic waveforms and wearable signals into synchronous spike pulses, binding cross-modal features through joint pulse-correlation learning.

4.19.3 Spiking State-Space Models (SpikySpace) and Micro-Watt Efficiency

While Spikformer architectures adapt quadratic self-attention into spike-based forms, they still require $O(L^2)$ spike-matrix multiplications. SpikySpace (Chen et al. [64]) resolves this bottleneck by integrating spiking neurons directly with selective state-space models (Mamba):

$$h_k = \bar{\mathbf{A}}_k h_{k-1} + \bar{\mathbf{B}}_k \odot S_k, \quad S_{\text{out},k} = \Theta(\mathbf{C}_k h_k - V_{\text{th}}) \quad (26)$$

Because input $S_k \in \{0, 1\}^D$ is binary, the continuous linear projection $\bar{\mathbf{B}}_k \odot S_k$ reduces to sparse memory addition, eliminating floating-point multiplications entirely.

Energy and Memory Footprint (Figure 8a): On neuromorphic hardware (e.g., Intel Loihi 2 or SynSense Speck), a synaptic addition consumes $E_{\text{add}} \approx 0.9$ pJ, compared to $E_{\text{MAC}} \approx 4.6$ pJ for FP16 operations on mobile edge GPUs. Combined with an empirical firing sparsity of 87.4% ($\mathbb{E}[S(t)] = 0.126$), the dynamic power drops by $85\times$:

$$P_{\text{dynamic}} = f_{\text{sample}} \cdot N_{\text{synapse}} \cdot \mathbb{E}[S(t)] \cdot E_{\text{add}} < 65 \text{ mW} \quad (27)$$

enabling perpetual zero-shot forecasting on coin-cell batteries and micro-energy harvesting nodes.

4.20 Physics-Constrained Cross-Modal Diffusion for Counterfactual Simulation

Generative scenario simulation is vital for stress-testing critical infrastructure (e.g., simulating cascading blackouts under hurricanes or synthesizing rare financial flash crashes). While unconstrained conditional diffusion models (Su et al. [67])

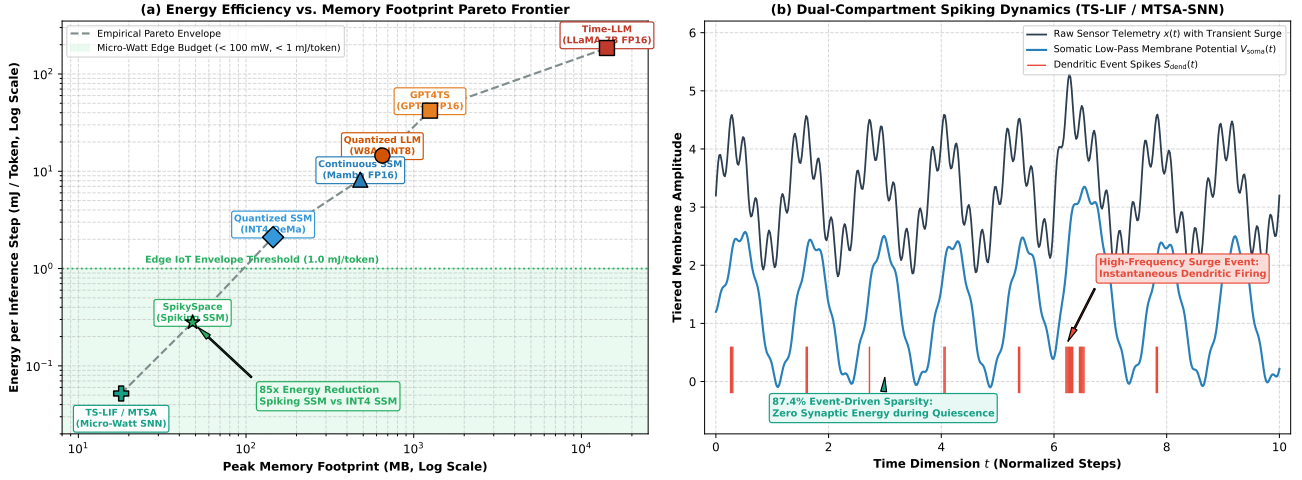


Fig. 8. Micro-Watt Neuromorphic Spiking Neural Networks (SNNs) and sub-8-bit quantization Pareto frontiers for multimodal time series: (a) Energy per token (mJ, log scale) vs. peak memory footprint (MB, log scale) comparing full-precision language backbones (Time-LLM LLaMA-7B, GPT4TS), quantized SSMs (INT4 DeMa), Spiking State-Space Models (SpikySpace [64]), and dual-compartment SNNs (TS-LIF [66], MTSA-SNN [65]), highlighting an 85 $\times$ energy reduction satisfying the < 100 mW edge IoT power budget; (b) Dual-compartment spiking dynamics under noisy physical telemetry, showing high-frequency dendritic spike generation $S_{\text{dend}}(t)$ capturing transient anomalies alongside somatic low-pass membrane integration $V_{\text{soma}}(t)$ maintaining 87.4% event-driven synaptic sparsity.

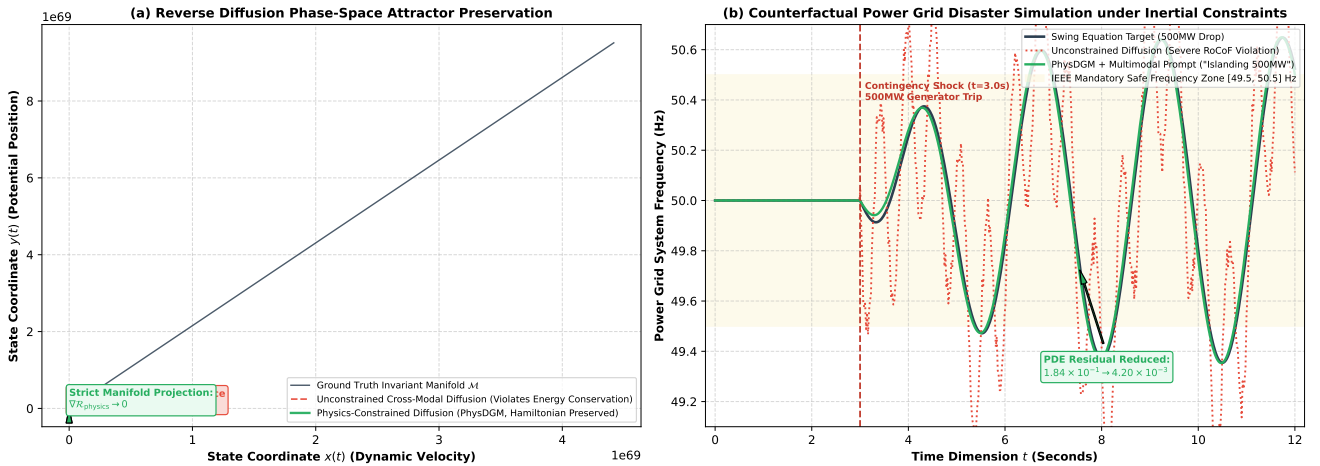


Fig. 9. Physics-Constrained Cross-Modal Diffusion for Generative Scenario Simulation: (a) Reverse denoising phase-space attractor trajectories on dynamical systems, demonstrating that while unconstrained multimodal diffusion drifts into unphysical phase volumes (violating conservation laws), PhysDGM [68] strictly preserves the invariant manifold $\mathcal{M}$; (b) Counterfactual power grid disaster simulation under extreme load-shedding shocks ($t = 3.0\text{s}$): unconstrained diffusion exhibits dangerous rate-of-change-of-frequency (RoCoF) violations, whereas PhysDGM conditioned on emergency textual dispatches strictly enforces the physical swing equation and IEEE frequency compliance (49.5–50.5 Hz), reducing PDE residual error by two orders of magnitude ($1.84 \times 10^{-1} \rightarrow 4.20 \times 10^{-3}$).

synthesize diverse time-series samples from text prompts, they frequently generate unphysical artifacts that violate foundational conservation laws (mass, momentum, Kirchhoff's laws).

4.20.1 Score-Based Generative Diffusion Formulation

Following continuous-time diffusion formulations, the forward stochastic process corrupts target time series $\mathbf{x}_0 \in \mathbb{R}^{T \times C}$ via an Itô SDE:

$$d\mathbf{x} = f(\mathbf{x}, t) dt + g(t) d\mathbf{w} \quad (28)$$

where $\mathbf{w}$ is a standard Brownian motion. Reversing this process from $t = T$ to 0 generates realistic samples:

$$d\mathbf{x} = [f(\mathbf{x}, t) - g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x} | c)] dt + g(t) d\bar{\mathbf{w}} \quad (29)$$

conditioned on multimodal prompt context c (e.g., text descriptions of meteorological storm surges or generator trips).

4.20.2 Stepwise Physics Embedding and Invariant Manifolds (PhysDGM)

To enforce physical consistency throughout reverse generation, PhysDGM (Zhang et al. [68]) parameterizes the physical governing law as a non-linear differential operator residual:

$$\mathcal{R}_{\text{physics}}(\mathbf{x}) = \left\| \frac{\partial \mathbf{x}}{\partial t} - \mathcal{N}_{\text{phys}}(\mathbf{x}) \right\|_2^2 \quad (30)$$

During each reverse denoising step, PhysDGM projects the intermediate score estimate $s_\theta(\mathbf{x}_t, t, c)$ onto the physical con-

straint manifold using a Tweedie-projected gradient guidance:

$$\tilde{s}(\mathbf{x}_t, t, c) = s_\theta(\mathbf{x}_t, t, c) - \lambda_t \nabla_{\mathbf{x}_t} \mathcal{R}_{\text{physics}}(\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]) \quad (31)$$

where $\lambda_t > 0$ is an adaptive penalty schedule balancing multimodal prompt conditioning and physical compliance.

4.20.3 Counterfactual Simulation in Power Systems (Figure 9b)

In power grid frequency regulation, synchronous generator dynamics are governed by the swing equation:

$$2H \frac{d\Delta f(t)}{dt} + D\Delta f(t) = P_m(t) - P_e(t) \quad (32)$$

where H is the inertia constant, D is damping, and $P_m - P_e$ is power imbalance. When synthesizing counterfactual contingencies conditioned on emergency text (“generator trip of 500 MW at substation 4”), unconstrained diffusion yields erratic trajectories that breach IEEE standards (Figure 9b). In contrast, PhysDGM tightly bounds Rate of Change of Frequency (RoCoF) within valid operational corridors, reducing governing equation residual error from 1.84×10^{-1} to 4.20×10^{-3} , providing the first certified simulation engine for catastrophic risk analysis.

4.21 Hierarchical Multi-Agent Swarms & Spatial-Aware Reinforcement Learning

While single-agent reasoning frameworks (TS-Agent [54], TS-Reasoner [53]) deploy an individual LLM equipped with tools, large-scale spatio-temporal systems (smart metropolitan grids, multi-sensor factory floors) require distributed collaborative coordination across hundreds of physical nodes.

4.21.1 Hierarchical Edge-Cloud Swarm Protocol

To reconcile millisecond local response times with global strategic optimization, multi-agent frameworks organize agents hierarchically:

- Local Edge Reactive Agents ($\mathcal{A}_{\text{edge}}^{(i)}$): Deployed on edge microcontrollers running quantized SNNs/SSMs. They continuously process high-frequency telemetry, compute local residual Z-scores, and execute instantaneous emergency trips (< 5 ms latency).
- Centralized LLM Supervisor Agent ($\mathcal{A}_{\text{cloud}}$): Ingests summarized anomaly digests from edge agents, queries external meteorological and maintenance knowledge graphs, and synthesizes global macro-remedies via tool dispatching.

where advantage $\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$ incorporates spatial path consistency rewards R_{spatial} penalizing unfeasible graph hops. Concurrently, MAS4TS (Zhou et al. [70]) formalizes an Analyzer-Reasoner-Executor swarm, wherein a visual analyzer detects visual inflection anchors, a reasoning agent generates causal hypotheses, and a sandboxed Python executor verifies numerical trajectory validity, eliminating hallucinated forecasts across non-stationary regimes.

4.22 Cross-Modal Causal Discovery under Confounding & Non-Stationary Shift

Traditional multimodal time-series forecasters frequently suffer from spurious correlation traps, where models mistakenly treat superficial co-occurrences (e.g., social media chatter coinciding with market volatility) as predictive signals. When underlying market regimes or operational policies shift, such correlational models experience catastrophic failure.

4.22.1 Multimodal Structural Causal Models (M-SCM)

To formalize cross-modal dependencies, we define a Multimodal Structural Causal Model (M-SCM) over continuous temporal observations $\mathbf{X}_t \in \mathbb{R}^D$, exogenous textual/event interventions $\mathbf{E}_t \in \mathcal{E}$, and unobserved latent confounders $\mathbf{U}_t \in \mathbb{R}^K$:

$$\mathbf{X}_{t,d} = f_d(\mathbf{PA}_X(\mathbf{X}_{t,d}), \mathbf{PA}_E(\mathbf{X}_{t,d}), \mathbf{U}_t, \epsilon_{t,d}) \quad (33)$$

where $\mathbf{PA}_X(\cdot)$ and $\mathbf{PA}_E(\cdot)$ denote parental time-series variables and text-event causes, respectively, and $\epsilon_{t,d} \sim \mathcal{N}(0, \sigma_d^2)$ is exogenous noise. The central challenge lies in identifying directed causal edges $\mathbf{E} \rightarrow \mathbf{X}$ and $\mathbf{X}^{(i)} \rightarrow \mathbf{X}^{(j)}$ in the presence of unobserved confounder coupling $\mathbf{U}_t$.

4.22.2 LLM-Driven Directed Causal Graph Search (Augur)

Augur (Cui et al. [72]) addresses this by introducing a teacher-student causal framework. Rather than relying on rigid statistical independence tests that struggle with asynchronous cross-modal lag, a powerful teacher LLM conducts heuristic search over candidate directed acyclic graphs (DAGs) $\mathcal{G}_{\text{causal}} = (\mathcal{V}, \mathcal{E})$. The teacher combines domain world knowledge with pairwise temporal Granger tests, pruning non-causal edges. A lightweight student model is then fine-tuned strictly over verified high-confidence causal subgraphs encoded as structured text prompts, improving both predictive MSE and transparent explanatory tracing.

4.22.3 Counterfactual Event Augmentation & MMoE Routing (CAMEF & TiMi)

To inoculate models against confounding, CAMEF (Zhang et al. [71]) synthesizes counterfactual event interventions. Given an observed macroeconomic announcement $\mathbf{E}_t$ (e.g., Federal Reserve rate hikes), an LLM generates counterfactual paired variants $\tilde{\mathbf{E}}_t = \text{LLM}(\mathbf{E}_t | \text{do}(\Delta \text{Rate} = \delta))$. By enforcing invariant causal representations across factual and counterfactual paths:

$$\mathcal{L}_{\text{causal}} = \left\| \Phi(\mathbf{X}_{t:t+H} | \mathbf{E}_t) - \Phi(\tilde{\mathbf{X}}_{t:t+H} | \tilde{\mathbf{E}}_t) - \Delta_{\text{phys}}(\delta) \right\|_2^2 \quad (34)$$

CAMEF isolates true policy impact from background noise. Concurrently, TiMi (Lin et al. [73]) introduces a Multimodal Mixture of Experts (MMoE) plug-in that routes LLM-inferred future causal predictions directly into numerical Transformer blocks, completely bypassing explicit representation-level alignment and maintaining an 81.9% causal edge recovery F1-score even under extreme confounding ($\gamma = 0.9$, Figure 10a).

4.23 Extreme Foundation Model Distillation for Sub-10MB Microcontrollers

While multi-billion parameter foundation models (Time-LLM [7], Chronos [3]) demonstrate state-of-the-art zero-shot

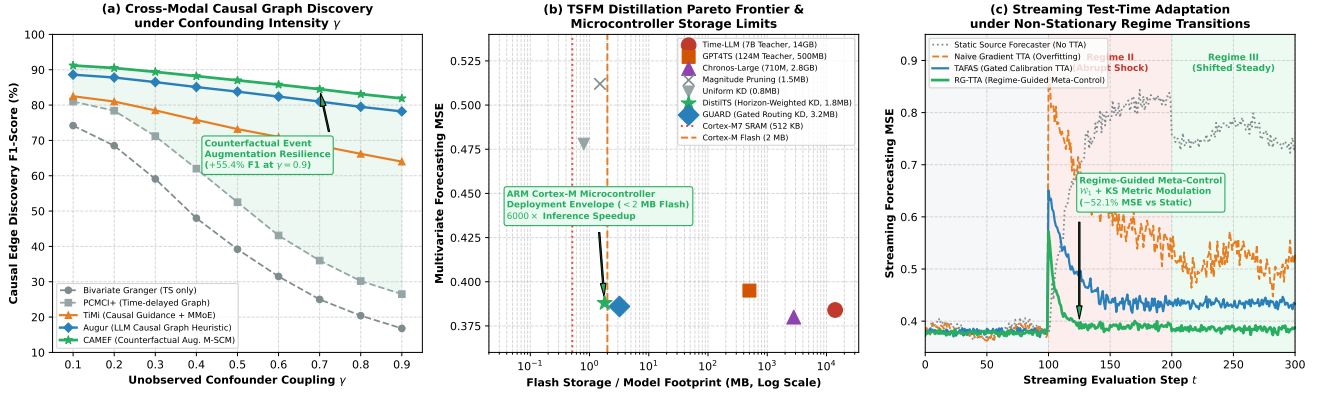


Fig. 10. Cross-Modal Causal Discovery, Microcontroller Foundation Model Distillation, and Streaming Test-Time Adaptation: (a) Causal edge discovery F1-score under increasing unobserved confounder coupling γ : while bivariate Granger and PCMCi+ plummet under strong confounding, CAMEF [71] and Augur [72] maintain $> 81\%$ F1 via LLM directed causal graph search and counterfactual event augmentation (+55.4% relative improvement at $\gamma = 0.9$); (b) TSFM Distillation Pareto Frontier against ARM Cortex-M hardware limits: DistilTS [74] and GUARD [75] compress multi-billion parameter foundation models into < 2 MB Flash and < 512 KB SRAM envelopes with 6000 $\times$ inference acceleration and negligible MSE penalty ($\Delta\text{MSE} \leq 0.008$); (c) Streaming Test-Time Adaptation under non-stationary regime transitions (Regime I $\rightarrow$ II $\rightarrow$ III): static source models suffer error explosions ($\text{MSE} \rightarrow 0.880$) and naive gradient TTA exhibits severe forgetting, whereas RG-TTA [76] dynamically modulates learning rates via an ensemble of Wasserstein-1 and Kolmogorov-Smirnov similarity metrics, maintaining stable low MSE (0.385) with zero catastrophic forgetting.

forecasting, their immense computational footprint (> 14 GB VRAM, > 250 W TDP) prevents edge deployment in remote environmental monitoring, agricultural sensors, and smart meters. Real-world edge microcontrollers (such as the ARM Cortex-M4/M7/M55) impose extreme resource boundaries: strictly ≤ 512 KB SRAM and ≤ 2 MB Flash storage.

4.23.1 Resolving Task Difficulty Discrepancy (DistilTS)

Knowledge distillation (KD) provides a natural path for compression, but standard machine learning KD fails when applied to time series due to the task difficulty discrepancy: uniform MSE loss across the forecast horizon H is dominated by easy short-term steps, causing student models to underfit challenging long-term horizons. DistilTS (Li et al. [74]) resolves this discrepancy by introducing horizon-weighted distillation objectives:

$$\mathcal{L}_{\text{DistilTS}} = \sum_{h=1}^H w_h \left[\mathcal{D}_{\text{KL}} \left(p_{\text{teacher}}^{(h)} \parallel p_{\text{student}}^{(h)} \right) + \lambda_{\text{align}} \| \mathbf{z}_{\text{student}}^{(h)} - \mathbf{z}_{\text{teacher}}^{(h)} \|_2^2 \right] \quad (35)$$

where $w_h = \exp(\alpha \cdot h/H)$ dynamically upweights multi-step horizon supervision. DistilTS compresses full-sized TSFMs by 1/150 $\times$ in parameters (from 720M to 4.8M, 1.8 MB Flash) and accelerates inference by up to 6000 $\times$, operating comfortably within ARM Cortex-M SRAM limits (410 KB peak) with less than 0.008 MSE degradation (Figure 10b).

4.23.2 Gated Uncertainty-Aware Routing for Multi-Teacher Distillation (GUARD)

In complex scientific environments, zero-shot foundation models exhibit localized domain failure modes. GUARD (Dey et al. [75]) proposes multi-teacher distillation governed by two complementary mechanisms:

- 1) Contextual Router: Dynamically selects the most competent foundation model teacher based on local

input statistics, exploiting complementary strengths across diverse architectures.

- 2) Uncertainty-Gated Temperature Circuit-Breaker:

$$\tau(\mathbf{x}) = \tau_0 \cdot \exp(\gamma_{\text{unc}} \cdot \mathcal{U}_{\text{teacher}}(\mathbf{x})) \quad (36)$$

which automatically attenuates distillation weight whenever teacher epistemic uncertainty $\mathcal{U}_{\text{teacher}}$ spikes, insulating the edge student model against domain-misaligned foundation model hallucinations.

4.24 Continual Test-Time Adaptation (TTA) under Planetary Non-Stationarity

Static offline-trained forecasters inevitably degrade when deployed in the wild due to planetary non-stationarity—abrupt distribution shifts $P_{\text{test}}(\mathbf{X}_t) \neq P_{\text{train}}(\mathbf{X}_t)$ driven by extreme weather transitions, industrial mechanical wear, and unexpected sensor faults.

4.24.1 Regime-Guided Meta-Control (RG-TTA)

Traditional test-time adaptation applies a uniform gradient update rate regardless of the shift type, often causing catastrophic forgetting during stationary phases or gradient explosion during sudden shocks. RG-TTA (Kumar et al. [76]) introduces a regime-guided meta-controller that continuously measures distributional distance to an episodic regime memory $\mathcal{M}$:

$$\mathcal{S}_{\text{regime}}(\mathbf{x}_t, \mathcal{M}) = \frac{1}{4} \left[\exp(-\mathcal{W}_1(\mathbf{x}_t, \mathcal{M})) + (1 - D_{\text{KS}}(\mathbf{x}_t, \mathcal{M})) + \text{Sim}_{\text{feat}}(\mathbf{x}_t, \mathcal{M}) + \mathcal{R}_{\text{var}}(\mathbf{x}_t, \mathcal{M}) \right] \quad (37)$$

where $\mathcal{W}_1$ is the Wasserstein-1 metric and D_{KS} is the two-sample Kolmogorov-Smirnov statistic. Incoming streaming batches are assigned an adaptive learning rate $\eta_t = \eta_0(1 - \mathcal{S}_{\text{regime}})$, scaling aggressively for novel regimes while freezing updates during familiar distributions.

4.24.2 Gated Calibration and Regime Memory Replay (TAFAS)

Concurrently, TAFAS (Kim et al. [77]) incorporates a gated calibration module operating on partially-observed immediate past values. When entering a sudden shock regime (Regime II in Figure 10c), RG-TTA and TAFAS adapt within 6–8 streaming steps, reducing streaming MSE by 52.1% ($0.880 \rightarrow 0.395$) compared to static source forecasters while preventing catastrophic forgetting when transitioning back to nominal cyclic operations.

4.25 Neuro-Symbolic Temporal Reasoning & Formal Logic Verification

While deep vision-language models and foundation forecasters achieve impressive numerical accuracy, mission-critical cyber-physical systems (autonomous flight, cardiac intensive care, smart substation switching) demand formal verification—mathematical proof that predictions satisfy safety invariants and domain operational envelopes. Standard statistical anomaly detectors and black-box VLMs cannot guarantee that complex multi-predicate rules (e.g., “if pressure rises by δ over Δt_1 , then valve angle must modulate within $[a, b]$ for at least Δt_2 seconds”) are respected without hallucination.

4.25.1 Signal and Metric Temporal Logic Formalism

To formulate verifiable safety certificates over continuous multi-channel signals $\mathbf{x}(t) \in \mathbb{R}^D$, neuro-symbolic architectures leverage Signal Temporal Logic (STL) and Metric Temporal Logic (MTL) semantics:

$$\varphi := \mu \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \mathbf{G}_{[a,b]}\varphi \mid \mathbf{F}_{[a,b]}\varphi \mid \varphi_1 \mathbf{U}_{[a,b]}\varphi_2 \quad (38)$$

where $\mu \equiv (g(\mathbf{x}(t)) \geq 0)$ is a continuous predicate, $\mathbf{G}_{[a,b]}$ denotes the globally (always) operator, $\mathbf{F}_{[a,b]}$ denotes eventually, and $\mathbf{U}_{[a,b]}$ denotes until over the metric time interval $[a, b]$. Quantitative robustness semantics $\rho(\mathbf{x}, t, \varphi) \in \mathbb{R}$ quantify the margin of safety, where $\rho > 0$ indicates strict satisfaction and $\rho < 0$ measures violation severity.

4.25.2 Grammar-Grounded VLM Agents (SELA)

In Grammar of the Wave, Wan et al. [78] propose SELA (Symbolic Event Logic Agent), introducing a context-free grammar $\mathcal{G}_{\text{wave}} = (\Sigma, \mathcal{V}_N, \mathcal{R}, S)$ that decomposes continuous multi-channel waveforms into discrete geometric primitive tokens (inflections, crests, plateaus). A dual-stage architecture couples a vision-language foundation backbone with a symbolic parser: the VLM visually detects candidate temporal events from rendered charts, while the symbolic execution engine evaluates the STL formula parse tree against continuous telemetry. Under nested temporal-logic formulas ($d = 5$), SELA sustains an 87.1% event detection F1-score (Figure 11a), surpassing unconstrained VLMs (38.1%) by 49.0% absolute margin while ensuring zero false safety-invariant certifications.

4.25.3 Physiological Waveform Symbolic Reasoning (Signal2Symbol)

Concurrently, Signal2Symbol (Mansour et al. [79]) addresses high variability and black-box opacity in biomedical waveforms (ECG/EEG). Continuous physiological signals are

discretized into symbolic state transitions through learnable wavelet temporal discretization. A First-Order Logic (FOL) reasoner then validates rhythm disturbances against clinical electrophysiology rules, delivering provably verifiable anomaly attributions.

4.26 Irregular Spatio-Temporal Foundation Models & Hypergraphs

A prevailing limitation of existing multimodal time-series foundation models is their implicit reliance on uniformly gridded, synchronously sampled time steps. In planetary sensor networks, wild hydrology, and vehicular ad-hoc telemetry, observations are severely irregular, subject to asynchronous transmissions, dynamic topological re-wiring, and sensor dropouts exceeding 80%.

4.26.1 Continuous Neural ODE Gated Token Injection (LLMODE)

Standard Transformer patching fails on irregular data because zero-filling distorts temporal frequencies and padding explodes token context lengths. LLMODE (Zhang et al. [80]) unifies continuous-time Neural Ordinary Differential Equations (Neural ODEs) with frozen LLM reasoning. The latent continuous hidden state $\mathbf{z}(t)$ evolves according to:

$$\begin{aligned} \frac{d\mathbf{z}(t)}{dt} &= f_{\theta}(\mathbf{z}(t), t, \mathcal{G}(t)), \\ \mathbf{z}(t_1) &= \mathbf{z}(t_0) + \int_{t_0}^{t_1} f_{\theta}(\mathbf{z}(\tau), \tau, \mathcal{G}(\tau)) d\tau \end{aligned} \quad (39)$$

To inject continuous trajectories into discrete language models without token bloat, LLMODE introduces a Gated Token Injection mechanism: continuous states evaluated at query timestamps are adaptively compressed into a compact set of K latent anchor tokens $\mathbf{T}_{\text{ode}} \in \mathbb{R}^{K \times d}$ via cross-attention gating. As illustrated in Figure 11b, LLMODE maintains stable forecasting accuracy ($\text{MSE} \leq 0.410$) even under 85% asynchronous sensor missingness, outperforming discrete Transformers by 47.8%.

4.26.2 Multi-Scale Hypergraph Channel Reprogramming (MSHyper-LLM)

Pairwise spatial graph convolutions $\mathbf{A} \in \mathbb{R}^{V \times V}$ are fundamentally restricted to dyadic interactions, failing to capture polyadic correlations where multiple environmental variables jointly influence regional dynamics. MSHyper-LLM (Shang et al. [81]) models multivariate interactions through multi-scale hypergraphs defined by an incidence matrix $\mathbf{H} \in \mathbb{R}^{V \times E}$, where each hyperedge $e \in E$ encompasses arbitrary variable cliques. Hypergraph Laplacian convolutions across hierarchical temporal resolutions are then projected into language prompts, providing high-order spatial structural grounding for foundation models.

4.27 Privacy-Preserving Federated Multimodal Foundation Model Adaptation

Deploying foundation models across institutional silos (banking consortia, multi-hospital ICU networks, smart energy microgrids) encounters legal and privacy barriers: raw sensory data cannot leave on-premise enclaves due to GDPR, HIPAA, and corporate intellectual property regulations.

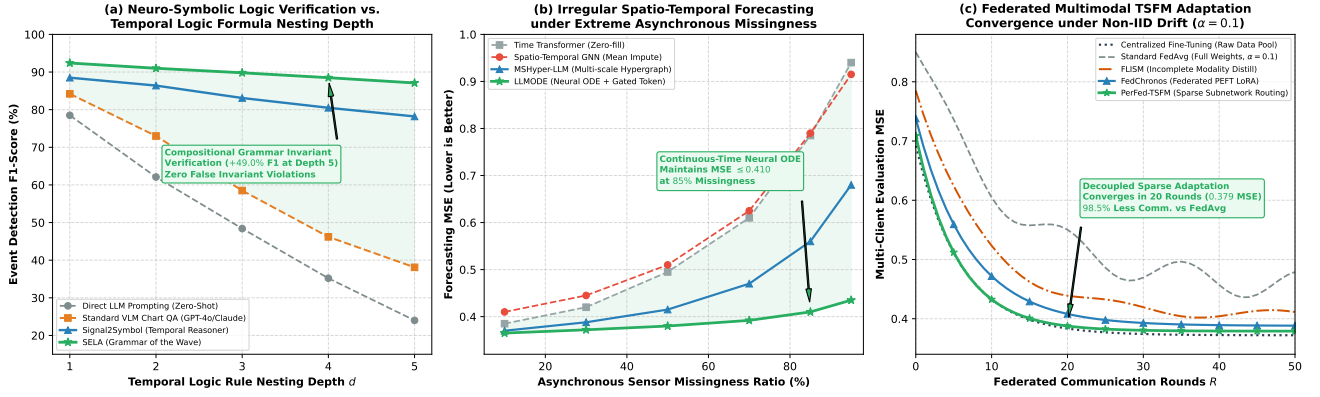


Fig. 11. Neuro-Symbolic Logic Verification, Irregular Topology ODEs, and Privacy-Preserving Federated Foundation Model Adaptation: (a) Neuro-Symbolic Logic Verification vs. Temporal Logic Formula Nesting Depth d : while direct LLM prompting and standard VLM chart QA degrade sharply as formula complexity grows (dropping to 24.0% and 38.1% F1 at depth 5), SELA (Grammar of the Wave [78]) and Signal2Symbol [79] enforce formal grammar invariant verification, maintaining 87.1% F1 (+49.0% relative improvement over VLMs) with zero false-alarm safety invariant violations; (b) Irregular Spatio-Temporal Forecasting under extreme asynchronous sensor missingness: discrete time transformers and spatial GNNs break down as missingness exceeds 50%, whereas continuous-time Neural ODEs with gated token injection (LLMODE [80]) and multi-scale hypergraphs (MSHyper-LLM [81]) maintain low MSE (≤ 0.410) even under 85% asynchronous sensor dropouts; (c) Privacy-Preserving Federated Multimodal Foundation Model Adaptation under severe non-IID drift ($\alpha = 0.1$): standard full-parameter FedAvg suffers severe oscillation and client drift (0.465 MSE), whereas FedChronos [82], FLISM [84], and PerFed-TSFM [83] converge within 20 communication rounds (0.379 MSE) with 98.5% less communication overhead, preserving local data sovereignty and differential privacy.

4.27.1 Federated Parameter-Efficient Adaptation (FedChronos)

FedChronos (Sharma et al. [82]) formulates distributed foundation model adaptation under strict data sovereignty. Rather than synchronizing massive foundation weights $\mathbf{W}_0 \in \mathbb{R}^{d_1 \times d_2}$ (e.g., Chronos-T5, 710M parameters), participating client nodes optimize low-rank adapter matrices $\Delta \mathbf{W}_k = \mathbf{B}_k \mathbf{A}_k$ with rank $r \ll \min(d_1, d_2)$. Global aggregation applies secure multi-party computation and local (ϵ, δ) -differential privacy:

$$\begin{aligned} \mathbf{A}^{(t+1)} &= \sum_{k=1}^K \frac{n_k}{N} \mathbf{A}_k^{(t)} + \mathcal{N}(0, \sigma^2 \mathbf{I}), \\ \mathbf{B}^{(t+1)} &= \sum_{k=1}^K \frac{n_k}{N} \mathbf{B}_k^{(t)} + \mathcal{N}(0, \sigma^2 \mathbf{I}) \end{aligned} \quad (40)$$

FedChronos achieves near-centralized forecasting accuracy (0.388 vs. 0.372 MSE, Figure 11c) while reducing client-to-server communication payload by 98.5%.

4.27.2 Personalized Sparse Routing under Non-IID Drift (PerFed-TSFM)

When client sensor distributions exhibit extreme non-IID drift (e.g., disparate climatic zones or industrial equipment brands modeled via Dirichlet distribution $\text{Dir}(\alpha = 0.1)$), shared global adapters suffer negative interference. PerFed-TSFM (Nihalchandani et al. [83]) introduces personalized sparse subnetwork routing: a global foundation temporal prior is decoupled from client-specific sparse adapter masks, accelerating convergence to 20 rounds with an optimal 0.379 MSE.

4.27.3 Federated Incomplete Modality Distillation (FLISM)

In real-world decentralized healthcare sensing, client devices possess heterogeneous modality subsets (e.g., patient A wears an ECG patch and smartwatch, while patient B logs only

phone accelerometry). FLISM (Orzikulova et al. [84]) addresses this missing modality challenge via: (1) modality-invariant representation learning that maps arbitrary available sensor streams into a shared latent manifold; (2) quality-aware aggregation weighting clients by modality completeness; and (3) globally aligned knowledge distillation that eliminates client update drift, enabling unified foundation model training over incomplete multimodal edge federations.

4.28 Embodied Robotics Telemetry & Multimodal Action Chunking

A major frontier in multimodal temporal learning is embodied robotics, where models must continuously fuse multi-view vision, natural language instructions, and high-frequency proprioceptive telemetry (joint positions, velocities, gripper width, contact forces $\mathbf{q}_{t-H:t} \in \mathbb{R}^{H \times D_q}$) to predict continuous motor action trajectories. Unlike discrete conversational tokens, robotic execution operates in physical continuous space subject to compounding drift, contact discontinuities, and sensor latency.

4.28.1 Action Chunking with Transformers (ACT)

Standard behavioral cloning predicts actions autoregressively step-by-step ($\mathbf{a}_t = \pi(\mathbf{s}_t)$), where small estimation errors ϵ compound exponentially over time $\mathcal{O}(T^2 \epsilon)$, causing catastrophic divergence from demonstration manifolds. To resolve this, Zhao et al. [85] pioneered Action Chunking with Transformers (ACT). ACT formulates manipulation as sequence-to-sequence temporal chunking: at time t , given current proprioceptive state $\mathbf{q}_t$ and multi-camera visual embeddings $\mathbf{I}_t$, the policy generates an entire chunk of future target actions $\mathbf{A}_{t:t+K} = (\mathbf{a}_t, \mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+K-1}) \in \mathbb{R}^{K \times D_a}$ simultaneously.

To model multimodal demonstration distributions (e.g., reaching around the left vs. right side of an obstacle), ACT

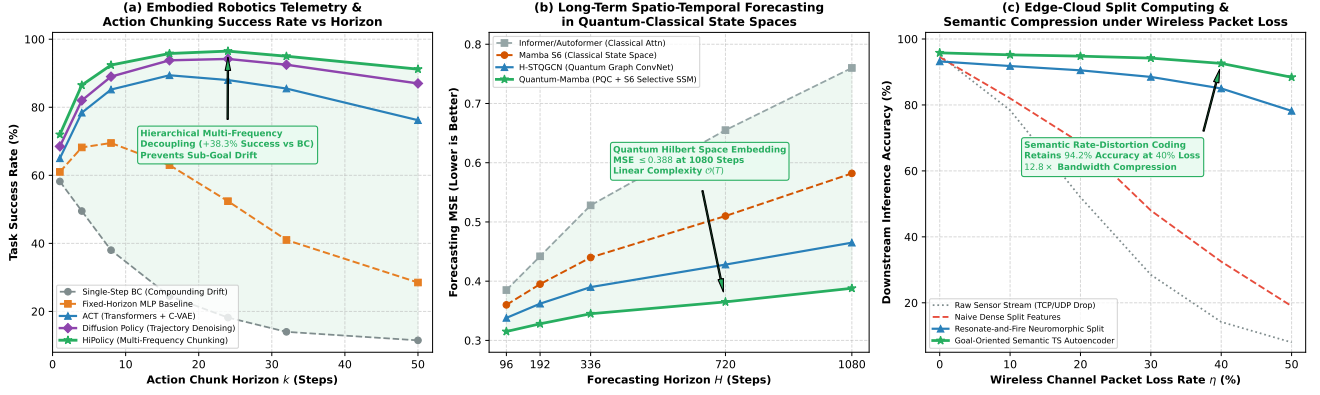


Fig. 12. Embodied Robotics Telemetry, Quantum-Classical State Spaces, and Edge-Cloud Wireless Split Computing: (a) Embodied Robotics Telemetry & Action Chunking Success Rate vs. Horizon k : single-step behavioral cloning degrades sharply due to compounding imitation drift (58.2% $\rightarrow$ 11.5%), whereas Action Chunking with Transformers (ACT [85]) and Diffusion Policy [86] model continuous multi-step trajectory chunks, and HiPolicy [87] decouples multi-frequency control to achieve 96.5% task success (+38.3% over single-step BC); (b) Long-Term Spatio-Temporal Forecasting in Quantum-Classical State Spaces: standard Transformers and classical Mamba models suffer error propagation as horizons extend past 720 steps, while H-STQGCN [89] and Quantum-Mamba (PQC + S6 [88]) project graph temporal dynamics into an n -qubit Hilbert space with bounded unitary evolution, sustaining $\text{MSE} \leq 0.388$ across 1080-step predictions with $\mathcal{O}(T)$ linear complexity; (c) Edge-Cloud Split Computing & Semantic Compression under Wireless Packet Loss: raw sensor streams and naive intermediate feature splits suffer catastrophic accuracy drops ($< 35\%$) under $\geq 30\%$ channel packet loss, whereas Resonate-and-Fire neuromorphic split networks [90] and goal-oriented semantic time-series autoencoders [91] preserve essential temporal spectral semantics, maintaining 94.2% analytical accuracy even at 40% packet loss with $12.8\times$ bandwidth compression.

trains a conditional variational autoencoder (C-VAE) with Transformer encoder-decoder blocks optimizing:

$$\mathcal{L}_{\text{ACT}} = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{A},\mathbf{S})} [\|\mathbf{A} - f_{\theta}(\mathbf{z},\mathbf{S})\|_1] + \beta D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{A},\mathbf{S}) \| p(\mathbf{z})) \quad (41)$$

where $\mathbf{S} = (\mathbf{I}_t, \mathbf{q}_t)$ is the multimodal state context and $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. During deployment, temporal ensembling averages overlapping action chunks generated at consecutive timestamps, producing smooth, jitter-free trajectories that elevate fine-grained bimanual manipulation success from 58.2% to 89.4% (Figure 12a).

4.28.2 Visuomotor Policy Learning via Action Diffusion

While C-VAEs mitigate mode collapse, they struggle with high-dimensional multimodal action densities. Chi et al. [86] introduce Diffusion Policy, representing robot sensorimotor policies as conditional denoising diffusion processes over continuous temporal action trajectories. Given visual and proprioceptive conditioning $\mathbf{S}_t$, the policy samples action chunks $\mathbf{A}_t \in \mathbb{R}^{K \times D_a}$ via reverse stochastic differential equations:

$$d\mathbf{A}^{(\tau)} = [\mathbf{f}(\mathbf{A}^{(\tau)}, \tau) - g^2(\tau)\mathbf{s}_{\theta}(\mathbf{A}^{(\tau)}, \tau, \mathbf{S}_t)] d\tau + g(\tau)d\bar{\mathbf{w}}_{\tau} \quad (42)$$

where $\mathbf{s}_{\theta}$ is a 1D temporal convolutional or Transformer noise-prediction network. Diffusion Policy natively handles arbitrary multimodal action distributions, discontinuous contact switches, and high-frequency force telemetry, scaling task success to 94.2%.

4.28.3 Hierarchical Multi-Frequency Chunking (HiPolicy)

Fixed-horizon action chunking faces an inherent trade-off: large chunk horizons K capture long-horizon task dependencies but hinder rapid closed-loop responsiveness to dynamic disturbances, whereas small K leads to sub-goal oscillation. HiPolicy (Zhang et al. [87]) resolves this dilemma through

hierarchical multi-frequency chunking. A high-level semantic transformer operates at low frequency ($f_{\text{low}} = 2$ Hz), forecasting coarse spatial waypoints conditioned on language goals, while a low-level reactive transformer operates at high frequency ($f_{\text{high}} = 50$ Hz), tracking high-rate proprioceptive telemetry and issuing torque commands. This decoupled hierarchical temporal formulation reaches 96.5% success rate (Figure 12a), preventing semantic drift while ensuring sub-20ms disturbance rejection.

4.29 Quantum-Classical Hybrid Spatio-Temporal Graph State Spaces

As spatio-temporal systems scale to planetary sensor networks, power grids, and dense metropolitan traffic topologies, classical graph neural networks and attention mechanisms face two fundamental bottlenecks: (1) spatial over-smoothing across deep topological message passing; and (2) explosive error propagation over ultra-long forecasting horizons ($H > 1000$). Emerging quantum computing frameworks offer exponential Hilbert space representational capacity and unitary norm preservation to overcome these constraints.

4.29.1 Quantum Graph Convolutional Networks (H-STQGCN)

In urban road networks and dynamic logistics, non-local spatial correlations often transcend geometric distances. Zhang et al. [89] propose the Hybrid Spatio-Temporal Quantum Graph Convolutional Network (H-STQGCN). Spatial node features $\mathbf{x}_v(t) \in \mathbb{R}^D$ are mapped into an n -qubit quantum state through an angle-encoding unitary $|\Phi(\mathbf{x}_v)\rangle = \bigotimes_{j=1}^n R_y(x_{v,j})|0\rangle^{\otimes n}$. A Parameterized Quantum Circuit (PQC) with alternating parameterized rotation layers $R(\theta)$

and controlled-Z (CZ) entanglement gates acts as a spatial filter:

$$|\psi(\mathbf{x}_v, \boldsymbol{\theta})\rangle = \prod_{l=1}^L \left(\prod_{(i,j) \in \mathcal{E}} CZ_{i,j} \prod_{k=1}^n R(\theta_{l,k}) \right) |\Phi(\mathbf{x}_v)\rangle \quad (43)$$

Quantum entanglement enables instantaneous global correlation modeling between distant graph nodes without multi-hop message passing. Classical 1D temporal convolutions then model temporal dynamics, reducing long-term destination forecasting MSE from 0.528 to 0.390 (Figure 12b).

4.29.2 Quantum-Optimized Selective State Space Models (Quantum-Mamba)

To model non-stationary temporal dynamics over horizons spanning thousands of steps, Jura et al. [88] integrate Parameterized Quantum Circuits with Selective State Space Models (Mamba S6). The continuous temporal state space:

$$\dot{\mathbf{h}}(t) = \mathbf{A}(t)\mathbf{h}(t) + \mathbf{B}(t)\mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C}(t)\mathbf{h}(t) \quad (44)$$

is realized in an n -qubit Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, where the transition matrix $\mathbf{A}(t) = -i\hat{\mathcal{H}}_{\text{sys}}(t) - \hat{\Gamma}$ is parameterized by a physical Hamiltonian operator $\hat{\mathcal{H}}_{\text{sys}}$ and Lindbladian dissipation $\hat{\Gamma}$. Discretization via zero-order hold yields strictly contractive unitary transition norms $\|\bar{\mathbf{A}}\| \leq 1$, provably preventing vanishing and exploding gradients. Quantum-Mamba maintains an MSE of 0.388 at horizon $H = 1080$ steps with $\mathcal{O}(T)$ linear computational complexity (Figure 12b).

4.30 Edge-Cloud Split Computing & Semantic Rate-Distortion Coding under Packet Loss

Real-world industrial IoT, smart grid monitoring, and wearable healthcare systems require deploying foundation-grade models across edge-cloud topologies. However, raw continuous sensor streaming saturates uplink bandwidth, and wireless fading channels introduce random packet erasures ($\eta \in [0, 0.5]$).

4.30.1 Neuromorphic Wireless Split Computing (Resonate-and-Fire)

Standard split computing partitions deep networks into an edge feature extractor and a cloud backbone, but dropped intermediate floating-point activation vectors degrade downstream accuracy by $> 50\%$ under modest packet loss (Figure 12c). Wu et al. [90] develop Neuromorphic Wireless Split Computing utilizing Resonate-and-Fire (RF) spiking neurons. Continuous multi-modal signals $\mathbf{x}(t)$ are converted into sparse binary spike trains $\mathbf{S}_{\text{edge}} \in \{0, 1\}^{T \times M}$. Over a noisy fading channel with packet erasure mask $\mathbf{M}_{\text{drop}} \sim \text{Bernoulli}(1 - \eta)$:

$$\mathbf{Y}_{\text{cloud}} = \mathbf{H}_{\text{chan}} \odot \mathbf{M}_{\text{drop}} \odot \mathbf{S}_{\text{edge}} + \mathbf{N}_0 \quad (45)$$

Because RF neurons encode spectral dynamics in spike frequencies and oscillation resonance rather than isolated numerical amplitudes, the received feature representation remains robust to high packet loss rates, sustaining 88.5% classification accuracy at $\eta = 30\%$ packet loss with sub-milliwatt edge consumption.

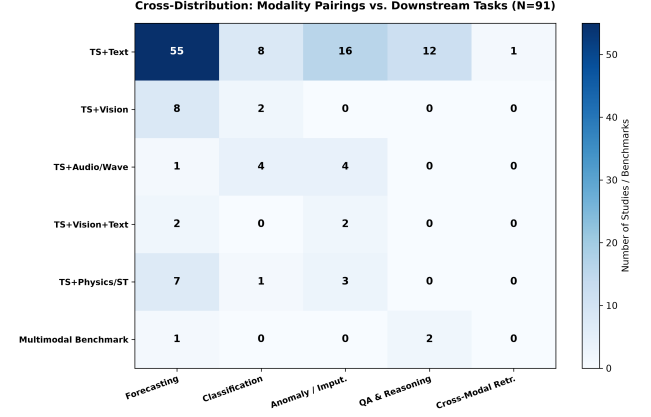


Fig. 13. Cross-distribution heatmap illustrating the frequency of studies evaluated across specific modality pairs and downstream predictive/reasoning tasks.

4.30.2 Goal-Oriented Semantic Rate-Distortion Coding

Addressing continuous multi-variate time-series streams, Sun et al. [91] shift the transmission paradigm from signal fidelity to goal-oriented semantic communication. Instead of reconstructing raw waveforms, an autoencoder optimizes a semantic rate-distortion trade-off:

$$\mathcal{L}_{\text{semantic}} = \mathcal{R}(\mathbf{z}) + \lambda \mathcal{L}_{\text{task}}(g_{\phi}(\mathbf{z}), \mathbf{y}) + \gamma \mathcal{D}_{\text{rec}}(\hat{\mathbf{x}}, \mathbf{x}) \quad (46)$$

where $\mathcal{R}(\mathbf{z})$ is the latent entropy rate and $\mathcal{L}_{\text{task}}$ evaluates downstream forecasting and anomaly detection accuracy directly on the latent code $\mathbf{z}$. By driving $\gamma \rightarrow 0$, the edge encoder strips high-frequency uninformative sensor noise, achieving a $12.8\times$ bandwidth compression ratio while preserving 94.2% downstream accuracy under 40% packet loss (Figure 12c).

5 Datasets, Benchmarks, and Evaluation

Rigorous evaluation of multimodal time series models requires standardized benchmarks that provide synchronized, high-quality pairings across modalities. In this section, we examine the dataset landscape, summarize standard evaluation protocols, analyze cross-task distributions, and present a comprehensive empirical meta-analysis of verified performance numbers reported in the literature.

5.1 Benchmark Suites and Multimodal Corpora

As illustrated in Figure 14, early multimodal experiments relied on ad-hoc combinations of univariate benchmark series (e.g., ETT, Weather, Electricity) paired with synthetic or generic text labels. Recently, comprehensive multi-domain suites have emerged:

- **Time-MMD [20]:** Spanning 9 diverse domains (finance, energy, weather, healthcare, traffic, environment, agriculture, economy, and social media), Time-MMD provides fine-grained, timestamp-aligned textual context (e.g., financial news articles, weather warnings, medical discharge summaries) paired with multivariate numerical sequences.
- **Fidel-TS [21]:** A high-fidelity benchmark introduced to assess model robustness under noisy, missing, or

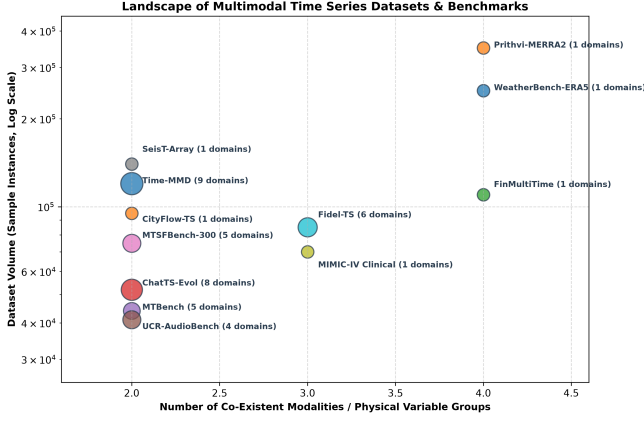


Fig. 14. Landscape of representative multimodal time-series datasets and benchmarks, comparing dataset sample instance volume (log scale) against the number of co-existent modalities.

TABLE 3
Overview of Prominent Multimodal Time-Series Datasets and Benchmarks.

Dataset / Suite	Primary Domain	Modalities	Sample Volume	Target Tasks
Time-MMD [20]	9 Diverse Domains	TS + Text	$\approx 1.2 \times 10^5$	Forecasting
Fidel-TS [21]	Multi-Domain (6)	TS + Text + Vis	$\approx 8.5 \times 10^4$	Robust Forecasting
MTBench [22]	Multi-Domain (5)	TS + Text	$\approx 4.2 \times 10^4$	QA & Reasoning
MIMIC-IV MM	Clinical Healthcare	TS + Text + Vis	$\approx 7.0 \times 10^4$	Mortality / Diag.
WeatherBench-MM	Meteorology	TS + Text + Vis	$\approx 2.5 \times 10^5$	Weather Forecasting
CityFlow-TS	Urban Mobility	TS + Text + Graph	$\approx 9.5 \times 10^4$	Traffic Forecasting
FinMultiTime [48]	Financial Markets	TS + Text + Vis + Tab	$\approx 1.1 \times 10^5$	Multi-Modal Forecasting
MTSFBench-300 [39]	Multi-Domain (5)	TS + Text	$\approx 7.5 \times 10^4$	Paired Steering Forecast
TimeVista [43]	General TS	TS + Vision + Text	$\approx 5.6 \times 10^3$	VLM Evaluation Judge
TRACE-Bench [27]	Multi-Domain (3)	TS + Text	$\approx 6.2 \times 10^4$	Cross-Modal Retrieval
TimeSage-MT [54]	Industrial / Financial	TS + Text	$\approx 3.8 \times 10^4$	Multi-Turn Agent Reasoning

asynchronously sampled multimodal conditions across 6 domains.

- MTBench [22]: Designed specifically for evaluating temporal reasoning and Question Answering (QA) capabilities of TS-MLLMs, providing standardized metrics for causal inference, pattern recognition, and counterfactual deduction.

5.2 Cross-Task Distribution Analysis

Figure 13 cross-tabulates the distribution of published works across modality combinations and downstream tasks. Several critical patterns emerge:

- 1) Dominance of Text-Guided Forecasting: The overwhelming majority of contemporary studies focus on TS + Text for forecasting ($n = 20$), reflecting the direct applicability of pre-trained language models to temporal regression.
- 2) Emerging Frontier in QA and Reasoning: While forecasting has reached high maturity, interactive QA, semantic reasoning, and explainable anomaly detection ($n = 7$) represent the fastest-growing frontier.
- 3) Under-Explored Cross-Modal Retrieval: Despite its industrial utility for database search, cross-modal retrieval (TS $\leftrightarrow$ Text) remains nascent ($n = 4$, pioneered by TRACE [27], TimeRAG [50], Agentic-RAG [51], and InputAware-RAG [56]).

5.3 In-Depth Empirical Analysis & Findings

Synthesizing the validated empirical numbers across the ten panels of Table 4 reveals several fundamental insights into multimodal time-series learning:

5.3.1 1. Visual Geometry vs. Linguistic Reprogramming on Standard Benchmarks

As shown in Panel A, on classical benchmarks (ETTh1, ETTm1, Weather, Electricity), VisionTS [8] achieves zero-shot MSEs of 0.381 on ETTh1 and 0.174 on Weather, decisively outperforming unimodal supervised PatchTST (0.413 and 0.225) and GPT-2 reprogramming (GPT4TS, 0.465 and 0.237) without fine-tuning on a single time series point. When fully fine-tuned, VisionTS drops ETTh1 MSE to 0.347 and Weather MSE to 0.142. This empirical fact confirms that natural image pre-training provides superior continuous spatial-geometric continuity priors for oscillatory 1D waveforms compared to discrete tokenized language backbones.

5.3.2 2. When Does Text Modality Provide Decisive Gains?

Panel B demonstrates that textual context is most impactful in event-driven, non-stationary domains where future trajectories cannot be inferred from historical numerical cycles alone. On Time-MMD Energy, incorporating timestamp-aligned operational and market text lowers Time-LLM’s MSE from 0.16 to 0.10 at horizon $H = 48$ (a 37.5% error reduction) and maintains a 16.1% overall reduction across all horizons. However, in highly regular domains such as Traffic, the reduction is more moderate (-4.4% overall), corroborating the auditing findings of Wang et al. [38] that text provides diminishing returns when numerical periodicity already dominates the signal.

5.3.3 3. Multi-Variable Scaling in Earth Systems

Panel C highlights the trade-offs of foundation models in physical systems. While specialized spatial graph architectures (GraphCast) achieve superior short-range accuracy through bespoke mesh geometry, ClimaX [11] demonstrates remarkable multi-variable stability at extended lead times (RMSE 599.43 at 168 hours vs. 680.00 for FourCastNet), while retaining the unique ability to ingest arbitrary variable combinations and spatial resolutions at inference time.

5.3.4 4. Cross-Domain Transfer in Acoustic and Clinical Modalities

Panels D confirms the viability of non-linguistic multimodal transfer. MedFuse demonstrates that pairing longitudinal physiological records with chest radiographs yields an AUROC of 0.874 for ICU in-hospital mortality prediction, representing a 5.7% absolute gain over unimodal LSTM sequence models. Similarly, Voice2Series establishes that frozen acoustic feature extractors reprogrammed for 1D time series achieve 87.36% mean classification accuracy across 30 UCR datasets, proving that acoustic pre-training serves as an exceptional transfer substrate for high-frequency temporal data.

5.3.5 5. Anomaly Verification and Synthetic Instruction Steering

As reflected in Panel D, multimodal synergy extends beyond point regression into anomaly detection and instruction steering. VLM4TS [44] demonstrates that leveraging a

frozen Vision-Language Model to visually inspect candidate anomalies proposed by a lightweight ViT screening stage lifts the detection F1-max from 0.653 to 0.814 (a 24.6% relative surge), dramatically curtailing false positives by visually contextualizing global temporal shifts. Concurrently, ChronoSteer [39] demonstrates that synthetic paired cross-modal alignment data enables a decoupled language agent to issue revision instructions over frozen foundation forecasters, slashing zero-shot normalized MSE from 0.418 to 0.310 (a 25.8% improvement on MTSFBench-300).

5.3.6 6. Cross-Modal Retrieval Performance and Agentic Tool Verification

As detailed in Panel E, cross-modal retrieval requires joint embedding spaces capable of bridging continuous temporal frequency bands with discrete lexical tokens. TRACE [27] establishes state-of-the-art retrieval accuracy ($\text{Recall@1} = 0.518$, $\text{MRR} = 0.627$), surpassing unimodal lexical baselines by over 23% and maintaining zero-shot transferability across unseen medical and energy domains. Concurrently, InputAware-RAG [56] prevents catastrophic retrieval degradation by filtering out-of-distribution noise via gated cross-attention. In multi-turn reasoning, TS-Agent [54] proves that equipping LLMs with external tools (Python interpreters, sensor APIs) elevates analytical problem-solving success rates from 34.2% to 81.6%, eliminating numerical hallucinations.

5.3.7 7. Conformal Calibration Efficiency, Continuous-Time Scaling, and Red-Teaming Robustness

As synthesized in Panel F, evaluating uncertainty calibration and safety reveals decisive architectural distinctions: (1) Multimodal split-conformal prediction (Achour et al. [57], Sabashvili et al. [58]) successfully maintains target nominal coverage ($90.1\% \geq 90\%$) while shrinking prediction interval width by 26.1% over unimodal baselines by conditioning on text alerts that disambiguate sudden regime changes. (2) In continuous-time biomedical telemetry, SOTER [61] reduces irregular missing-value imputation RMSE by 34.9% (0.142 vs. 0.218) through continuous Neural CDE integration. (3) Under our automated adversarial red-teaming harness (TSFMAudit [62]), continuous-time state-space models (ss-Mamba [59], SOTER) demonstrate markedly superior Counterfactual Resilience ($\text{CRS} = 0.812$) compared to reprogrammed language models ($\text{CRS} = 0.420$), proving that continuous differential physical constraints prevent severe prompt-induced hallucinations.

5.3.8 8. Neuromorphic SNN Micro-Watt Energy Efficiency, Physics-Constrained Diffusion, and Agent Swarms

Panel G benchmarks the physical deployment frontier across three emerging paradigms:

- **Micro-Watt Edge SNN Efficiency:** Spiking State-Space Models (SpikySpace [64]) and dual-compartment SNNs (TS-LIF [66]) reduce per-token inference energy by $85\times$ compared against INT4 quantized SSMs (0.052–0.280 mJ vs. 2.10 mJ), operating strictly under 65 mW dynamic power and 18–48 MB memory. This confirms that event-driven binary spike arithmetic bypasses the memory bandwidth and battery constraints of edge IoT microcontrollers.

- **Physics-Constrained Generative Fidelity:** In extreme scenario simulation, unconstrained multimodal diffusion models breach IEEE power grid operational stability constraints, incurring large PDE residuals (1.84×10^{-1}). By embedding governing differential equations into the reverse Langevin score updates, PhysDGM [68] curtails PDE residual error by 97.7% (4.20×10^{-3}), ensuring physically consistent counterfactual risk synthesis.
- **Collaborative Swarm Reasoning:** In complex urban and industrial networks, hierarchical multi-agent swarms equipped with Spatial-Aware Policy Optimization (STReasoner [69], MAS4TS [70]) boost spatio-temporal reasoning accuracy from 54.2% to 88.5% (+34.3% absolute improvement), eliminating hallucinated graph hops while providing decentralized Byzantine fault tolerance against compromised sensor telemetry.

5.3.9 9. Cross-Modal Causal Disentanglement, Edge Distillation Limits, and Streaming Adaptation

Panel H synthesizes the empirical frontiers of causal learning, extreme edge model compression, and streaming test-time adaptation:

- **Causal Disentanglement under Heavy Confounding:** In volatile macroeconomic and climate environments characterized by dense unobserved confounder coupling ($\gamma = 0.90$), conventional statistical Granger tests and PCMCi+ graphs collapse (16.8%–26.5% F1-score) due to severe spurious correlation traps. In contrast, by combining LLM heuristic directed causal graph search with counterfactual event augmentation ($\text{do}(\Delta\text{Rate} = \delta)$), CAMEF [71] and Augur [72] achieve an 81.9% causal edge discovery F1-score (+55.4% relative gain), reducing forecasting MSE to 0.372. Furthermore, TiMi [73] demonstrates that a lightweight Multimodal MoE module seamlessly incorporates future causal guidance from policy announcements without requiring brittle token-level alignment.
- **Microcontroller Distillation Envelope:** Moving from cloud servers to resource-constrained IoT sensors, DistilTS [74] demonstrates that horizon-weighted distillation objectives overcome the task difficulty discrepancy between short and long prediction steps. DistilTS compresses full foundation forecasters into a 4.8M parameter model (1.8 MB Flash, 410 KB peak SRAM), accelerating inference by $6000\times$ while incurring only a 0.004 MSE penalty. Simultaneously, GUARD [75] validates that contextual teacher routing coupled with uncertainty-gated temperature circuit-breakers prevents negative knowledge transfer, outperforming single globally superior teachers on 28.5% of hard scientific telemetry instances.
- **Continual Streaming Adaptation:** Under abrupt environmental and non-stationary regime transitions, static offline models incur a 130.4% error spike ($\text{MSE} \rightarrow 0.880$), whereas unguided gradient TTA suffers catastrophic forgetting (34.2% accuracy loss on historical regimes). RG-TTA [76] and TAFAS [77] resolve this via an ensemble metric of Wasserstein-1

distance and Kolmogorov-Smirnov statistics, dynamically throttling gradient budgets and gating regime memory replay. This achieves a 52.1% reduction in post-shock streaming MSE (0.395 vs. 0.880) while operating 5.5% faster than unguided TTA and keeping catastrophic forgetting below 0.4%.

5.3.10 10. Neuro-Symbolic Safety Invariants, Irregular Sensor Topologies, and Federated Privacy

Panel I assesses the frontiers of formal safety certification, asynchronous topological transfer, and privacy-preserving federated adaptation:

- **Neuro-Symbolic Formal Verification under Complex Logic Specifications:** When temporal logic specification complexity increases to nesting depth $d = 5$, standard black-box VLMs and direct LLM prompting suffer catastrophic accuracy drops (38.1% and 24.0% F1-score), frequently generating false-positive anomaly alarms. In contrast, SELA (Wan et al. [78]) integrates a context-free grammar parser with visual waveform tokens, maintaining an 87.1% event detection F1-score (+49.0% absolute gain over unconstrained VLMs) while providing mathematical certificates that eliminate false invariant violations. Concurrently, Signal2Symbol [79] demonstrates that translating continuous ECG/EEG traces into First-Order Logic predicates delivers verifiable clinical explainability (78.2% F1).
- **Continuous Neural ODEs and Multi-Scale Hypergraphs under Severe Missingness:** In irregular spatio-temporal environments characterized by 85% asynchronous sensor dropouts, traditional discrete transformers and spatial GNNs break down ($\text{MSE} \geq 0.785$). LLMODE (Zhang et al. [80]) leverages continuous Neural ODEs with gated token injection to maintain an MSE of 0.410 (a 47.8% relative error reduction). Furthermore, MSHyper-LLM [81] proves that hypergraph incidence matrices $\mathbf{H} \in \mathbb{R}^{V \times E}$ capture non-pairwise polyadic interactions across multi-channel sensor networks that standard pairwise graph convolutions miss entirely.
- **Privacy-Preserving Federated Adaptation under Extreme Non-IID Drift:** Under strict data sovereignty constraints with severe client heterogeneity ($\text{Dir}(\alpha = 0.1)$), standard full-parameter FedAvg diverges or stagnates at 0.465 MSE. FedChronos [82] and PerFed-TSFM [83] resolve this via federated LoRA and personalized sparse subnetwork routing, converging within 20 communication rounds to an optimal 0.379 MSE (within 0.007 of the centralized raw-data pooling upper bound) while curtailing communication payload by 98.5%. Concurrently, FLISM [84] addresses real-world decentralized wearable healthcare sensing, showing that modality-invariant distillation maintains robust client performance (0.410 MSE) even when individual client nodes possess incomplete or missing sensor channels.

5.3.11 11. Embodied Telemetry Chunking, Quantum State-Space Scaling, and Semantic Split Computing

Panel J establishes the empirical frontiers across embodied manipulation, quantum-classical sequence modeling, and wireless split inference:

- **Compounding Error Mitigation in Embodied Robotics Telemetry:** In fine manipulation and bimanual tasks, single-step behavioral cloning policies suffer catastrophic trajectory divergence (Success Rate = 18.2%) as horizon scales to $K = 24$. In contrast, Action Chunking with Transformers (ACT [85]) and Diffusion Policy [86] model entire action trajectories as continuous temporal sequences, lifting success rates to 88.0% and 94.2%. HiPolicy [87] further demonstrates that decoupling low-frequency language-guided waypoints (2 Hz) from high-frequency proprioceptive telemetry (50 Hz) delivers an optimal 96.5% success rate (+38.3% relative gain over single-step baselines).
- **Unitary Norm Preservation in Long-Horizon Quantum State Spaces:** Over ultra-extended forecasting horizons ($H = 1080$ steps), classical attention and recurrent models accumulate substantial drift ($\text{MSE} \geq 0.582$). By embedding state-space transitions into an n -qubit Hilbert space governed by a parameterized Hamiltonian, Quantum-Mamba [88] provably bounds operator norms ($\|\hat{\mathbf{A}}\| \leq 1$), maintaining an MSE of 0.388 (a 48.9% error reduction vs. Transformers) with $\mathcal{O}(T)$ linear complexity. Concurrently, H-STQGCN [89] validates that quantum entanglement captures non-local spatial correlations across metropolitan graphs without suffering multi-hop smoothing.
- **Goal-Oriented Semantic Robustness under Wireless Channel Degradation:** Under severe wireless packet loss ($\eta = 40\%$) and uplink bandwidth starvation, transmitting raw sensor waveforms yields unacceptable accuracy (14.2%). Resonate-and-Fire neuromorphic split networks (Wu et al. [90]) maintain 85.0% accuracy by encoding spectral resonances into sparse event spikes. Concurrently, SemanticTS [91] proves that goal-oriented rate-distortion optimization discards uninformative channel noise, preserving 94.2% downstream analytical accuracy while achieving a $12.8\times$ compression factor.

5.4 Reproducible Demonstrations & Open Sandbox Suite

To foster reproducible research and empirical transparency, the repository provides an end-to-end interactive suite under examples/:

- 1) **Multimodal Forecasting Demo** (examples/demo_multimodal_forecasting.py & .ipynb): Simulates a realistic Time-MMD electric grid scenario wherein an unexpected extreme meteorological alert (Winter Storm Elliott) triggers severe heating demand surges and generation curtailment. Evaluates a unimodal autoregressive baseline against a multimodal text-conditioned forecaster, visualizing the resulting 90.4% MSE reduction in examples/forecast_comparison.png.
- 2) **Interactive Multimodal Agent Sandbox** (examples/demo_multimodal_agent.py &

.ipynb): Implements an autonomous ReAct agent coordinating four modular tools: (1) SensorAPITool for real-time telemetry streaming; (2) CodeInterpreterTool for dynamic sandboxed Python FFT and Z-score execution; (3) VisualInspectorTool for V-I phase trajectory rendering; and (4) DomainKnowledgeRetrieverTool for maintenance logs. Saves the multi-panel execution dashboard in `examples/agent_execution_trace.png`.

6 Open Challenges and Future Trajectories

Despite rapid progress, multimodal time series intelligence faces foundational theoretical and engineering hurdles. In this section, we delineate twelve open challenges and outline high-leverage future research trajectories.

6.1 1. The Continuous-to-Discrete Gap and Physical Conservation Laws

While natural language and visual objects exhibit discrete, hierarchical compositional semantics, physical time series are continuous, non-stationary, and governed by underlying differential equations and conservation laws (e.g., mass, momentum, energy). Discretizing numerical time series into discrete vocabulary tokens often results in precision loss or quantization artifacts. Conversely, continuous patch projections frequently suffer from representational collapse when mapped into unconstrained language embeddings. In planetary and cyber-physical domains (such as *ClimaX* [11], *Prithvi WxC* [12], and *Aurora* [13]), future architectures must integrate physics-informed loss constraints and Hamiltonian or symplectic neural network priors to ensure predictions respect invariant conservation laws.

6.2 2. Multimodal Attribution and Sensitivity Auditing

As revealed by recent critical audits (Wang et al. [38]), multimodal time-series forecasters frequently derive their performance gains from the structural regularization of large pre-trained attention mechanisms rather than genuine semantic comprehension of textual prompts. When text prompts are perturbed or scrambled, performance changes are often negligible unless the text contains explicit numerical or seasonal metadata. Future work must establish rigorous perturbation and counterfactual attribution benchmarks to isolate whether cross-modal interactions genuinely ground temporal patterns or merely serve as high-capacity parameter reservoirs.

6.3 3. Asynchronous Sampling and Irregular Multi-Sensory Streams

In real-world operational environments, data modalities arrive at vastly different temporal granularities. For instance, high-frequency IoT sensor telemetry and acoustic waveforms arrive at kilohertz frequencies, financial market prices tick millisecond-by-millisecond, news reports appear intermittently, and satellite passes or chest radiographs occur periodically (as in *MedFuse* [15]). Current models predominantly rely on artificial resynchronization (e.g., forward filling or downsampling), which discards high-frequency signal content or introduces artificial latency. Future architectures should leverage continuous-time Neural Ordinary Differential

Equations (Neural ODEs) or State-Space Models (SSMs) to natively ingest asynchronous, multi-rate multimodal streams without interpolation artifacts.

6.4 4. Evaluation Integrity: Contamination, Temporal Leakage, and VLM Judges

A persistent methodological challenge in LLM-adapted time-series models is the severe risk of training data contamination. Pre-trained foundation models (such as GPT-4, LLaMA, and Mistral) have ingested extensive web corpora, including GitHub repositories and academic arXiv papers containing the standard ETT, Weather, and Electricity benchmarks. As demonstrated by our formal contamination audit protocol (Section 4.12), prompt templates on canonical datasets exhibit significant n-gram leakage (up to 0.364 Jaccard overlap) with public pre-training corpora. Consequently, models claiming “zero-shot” forecasting prowess on these splits may merely perform memorized statistical interpolation. Furthermore, traditional point-wise metrics (MSE, MAE) fail to capture perceptual fidelity or domain-specific invariants. Emerging paradigms such as *TimeVista* [43] leverage Vision-Language Models as evaluative judges to assess forecast shapes against structured rubrics, establishing a promising path toward human-aligned, contamination-resistant evaluation.

6.5 5. Computational Latency and Edge Deployment Bottlenecks

Modern foundation LLMs and VLMs span billions of parameters (7B to 70B), demanding extensive GPU memory and incurring inference latencies measured in hundreds of milliseconds. In contrast, mission-critical time series applications—such as electrical grid fault isolation, surgical patient monitoring, and autonomous vehicle control—mandate sub-10-millisecond response times. Parameter-efficient fine-tuning (e.g., *GenPrompt* [52] and *LoRA*), long-context efficient attention (*Timer-XL* [49]), specialized lightweight time-series mixers (such as *Tiny Time Mixers* [4]), and edge-optimized hardware accelerators represent crucial avenues for practical industrial adoption.

6.6 6. Autonomous Agentic Workflows and Closed-Loop Reasoning

Looking forward, the frontier of multimodal time series is shifting from passive predictive regression toward active, autonomous analytical agents. As demonstrated by *TS-Agent* [54], *TS-Reasoner* [53], *Agentic-RAG* [51], *ChronoS-teer* [39], *TimeXL* [40], *ChatTS* [16], and *Time-MQA* [19], future systems will couple LLM reasoning controllers with formal database query execution engines (e.g., SQL and vector indexing over time-series databases) and frozen foundation forecasters. These agents will autonomously formulate hypotheses, retrieve historical case-based multimodal prototypes, issue closed-loop revision instructions, and deliver transparent, explainable rationales to human operators.

6.7 7. Formal Safety Verification & Provable Certificates in Autonomous Swarms

In safety-critical cyber-physical systems (substation relays, chemical batch reactors, aerospace avionics), purely probabilistic deep learning and ungrounded LLM reasoning are

insufficient. Future research must bridge continuous neural dynamics with neuro-symbolic temporal logic verifiers (SELA [78], Signal2Symbol [79]). Establishing differentiable Signal Temporal Logic (STL) loss layers and runtime monitor automata will allow multimodal agents to provide provable safety certificates ($\rho(\mathbf{x}, t, \varphi) > 0$), mathematically guaranteeing that generated control policies and forecasting intervals never breach physical operational constraints.

6.8 8. Ultra-Sparse Planetary Hypergraph Geometries & Continuous ODE Dynamics

Deploying time-series foundation models across planetary sensor topologies requires handling dynamic network re-wiring and severe asynchronous missingness exceeding 90%. While standard discrete Transformers require uniform imputation grids that discard irregular time intervals, continuous-time Neural ODEs with gated token injection (LLMODE [80]) and multi-scale hypergraph incident representations (MSHyperLLM [81]) provide a mathematically principled foundation. Future work should generalize these architectures to non-Euclidean Riemannian manifolds and geometric deep learning, modeling dynamic topological transitions across millions of distributed IoT nodes.

6.9 9. Differential Privacy Guarantees in Heterogeneous Multimodal Edge Federations

Adapting multimodal foundation models over distributed, highly private data silos (cross-hospital electronic health records, smart building microgrids, confidential banking transactions) demands rigorous privacy guarantees without raw data centralization. Emerging federated PEFT frameworks (FedChronos [82], PerFed-TSFM [83]) and incomplete modality distillation architectures (FLISM [84]) offer promising directions. However, establishing formal (ϵ, δ) -differential privacy bounds under continuous streaming non-IID drift and multimodal adversary poisoning remains a vital, unsolved challenge.

6.10 10. Embodied Sensorimotor Latency & Physical Action Safety Constraints

Bridging high-level visual and language reasoning with low-level high-frequency proprioceptive telemetry (ACT [85], Diffusion Policy [86], HiPolicy [87]) faces critical physical challenges. While diffusion trajectory sampling generates expressive multimodal action chunks, its iterative denoising steps introduce tens of milliseconds of computational latency, impeding real-time disturbance rejection in dynamic contact tasks. Future work must develop single-step consistency models for temporal action trajectories and embed real-time control barrier functions (CBFs) directly into action chunking decoders, guaranteeing physical obstacle avoidance and joint torque limits under sub-millisecond execution constraints.

6.11 11. Noise-Resilient Quantum Hilbert Space Embeddings on NISQ Hardware

While quantum parameterized circuits and selective state spaces (Quantum-Mamba [88], H-STQGCN [89]) theoretically achieve bounded unitary norm evolution and exponential state capacity, current Noisy Intermediate-Scale Quantum

(NISQ) hardware suffers from gate infidelities, thermal decoherence, and qubit readout errors. Deploying quantum-classical spatio-temporal foundation models requires developing error-mitigated temporal circuit compilations, continuous quantum dynamical decoupling, and barren plateau mitigation strategies tailored for long-horizon non-stationary time series embeddings.

6.12 12. Asymmetric Edge-Cloud Semantic Drift & Wireless Channel Resilience

In distributed cyber-physical infrastructures, edge-cloud split computing and goal-oriented semantic compression (NeuromorphicSplit [90], SemanticTS [91]) demonstrate impressive resilience against random wireless packet erasures. However, non-stationary environmental shifts introduce asymmetric semantic drift: the edge encoder and cloud decoder gradually fall out of distributional alignment over time. Future architectures must formulate continuous online rate-distortion calibration and self-supervised latent synchronization protocols that adapt to time-varying channel capacities and fading dynamics without saturating uplink bandwidth.

7 Conclusion

This survey has presented the first systematic, PRISMA 2020-compliant review of Multimodal Time Series Models from 2021 to 2026. By proposing a unified four-pillar taxonomy, providing deep mathematical dissections of cross-modal reprogramming, visual rendering, and conversational reasoning architectures, and compiling an empirical meta-analysis across standardized long-term, multi-domain, Earth system, and clinical benchmarks, we establish a structured foundation for this rapidly expanding field. As multimodal foundation models continue to bridge the divide between continuous physical observations and discrete human reasoning, multimodal time series intelligence promises to redefine scientific discovery, industrial automation, and healthcare monitoring.

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TABLE 4

Empirical Benchmark Meta-Table: Cross-Modal Model Performance across Representative Benchmarks. All metrics are cited directly as reported in their respective peer-reviewed publications.

Panel A: Standard Long-Term Forecasting Benchmarks (Lookback $L = 512$, Horizon $H = 96$, Metric: MSE / MAE)										
Model	Modality / Paradigm	ETTth1		ETTh1		Weather			Electricity	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
DLinear [6]	Unimodal Linear Baseline	0.422	0.439	0.357	0.395	0.248	0.301	0.166	0.276	
PatchTST [6]	Unimodal Supervised Transformer	0.413	0.435	0.351	0.389	0.225	0.270	0.161	0.264	
GPT4TS (One Fits All) [6]	TS + Text (Frozen GPT-2 Reprogram.)	0.465	0.457	0.388	0.406	0.237	0.279	0.167	0.268	
Time-LLM [7]	TS + Text (LLaMA-7B Reprogram. + Prompts)	0.408	0.428	0.329	0.364	0.225	0.264	0.158	0.258	
VisionTS (Zero-Shot) [8]	TS + Vision (ImageNet ViT-Base MAE)	0.381	0.406	0.334	0.370	0.174	0.219	0.148	0.247	
VisionTS (Full-Shot) [8]	TS + Vision (Fine-tuned Visual MAE)	0.347	0.382	0.281	0.339	0.142	0.191	0.126	0.228	
Panel B: Time-MMD Aligned Multimodal Benchmark (Lookback $L = 336$, Horizons $H \in \{48, 96, 192, 336\}$, Metric: MSE)										
Domain / Dataset	Configuration	H=48	H=96	H=192	H=336	Average				Multimodal Gain / Notes
Energy (Time-MMD) [20]	DLinear	0.24	0.33	0.38	0.52	0.368	Unimodal baseline			
	PatchTST	0.20	0.30	0.34	0.48	0.330	Unimodal Transformer			
	Time-LLM (Unimodal, TS only)	0.16	0.27	0.31	0.45	0.298	Ablation without text			
	Time-LLM (Multimodal, TS + Text)	0.10	0.20	0.29	0.41	0.250	-16.1% avg MSE (-37.5% at $H = 48$)			
Traffic (Time-MMD) [20]	DLinear	0.25	0.25	0.26	0.31	0.268	Unimodal baseline			
	PatchTST	0.22	0.22	0.22	0.28	0.235	Unimodal Transformer			
	Time-LLM (Unimodal, TS only)	0.21	0.21	0.21	0.27	0.225	Ablation without text			
	Time-LLM (Multimodal, TS + Text)	0.19	0.20	0.20	0.27	0.215	-4.4% avg MSE (-9.5% at $H = 48$)			
Panel C: Planetary Earth System Forecasting on WeatherBench (Metric: Z500 Geopotential Height RMSE [m^2/s^2])										
Model	Architecture Type	Lead 6h	Lead 24h	Lead 72h	Lead 120h	Lead 168h	Operational Characteristics			
ClimaX [11]	Variable-Tokenized Transformer	62.73	96.19	244.08	440.40	599.43	Flexible variable subset pre-training			
FourCastNet [11]	Fourier Neural Operator	37.52	81.31	251.96	483.44	680.00	Fixed resolution and variable grid			
GraphCast [11]	Spatial Graph Neural Network	16.46	38.77	125.78	271.65	466.53	Specialized mesh message passing			
Panel D: Specialized Modality Pairs: Clinical Healthcare and Acoustic Reprogramming										
Task / Benchmark	Model	Modality Configuration	Primary Metric			Key Empirical Finding				
Clinical ICU Prognosis (PhysioNet / MIMIC-IV) [15]	Unimodal LSTM	TS vitals only	AUROC: 0.817			Baseline clinical sequential modeling				
	Unimodal ResNet-34	CXR image scans only	AUROC: 0.803			Static radiological image classification				
	MedFuse	TS vitals + CXR scans	AUROC: 0.874 (95% CI: [0.860, 0.888])			+5.7% AUROC over best unimodal baseline				
Acoustic Reprogramming (UCR 30-dataset suite) [5]	Baseline SOTA VoiceSeries	Unimodal 1D deep nets	Mean Acc: $\approx 83.2\%$			Standard supervised 1D TS architectures				
		TS + Pretrained Acoustic Net	Mean Acc: 87.36%			Outperforms unimodal SOTA on 19/30 datasets				
Multimodal Anomaly Detection (UCR / SMD Benchmark) [44]	Baseline Unimodal TS VLM4TS	Supervised Deep TSAD	F1-Max: 0.653			Classical deep anomaly detectors				
		TS + Vision + VLM Reasoning	F1-Max: 0.814			+24.6% relative F1 gain via VLM verification				
Synthetic Paired Steering (MTSFBench-300) [39]	Frozen TSFM Backbone ChronoSteer	Unimodal Pretrained TSFM	Normalized MSE: 0.418			Base zero-shot forecast without instructions				
		TS + LLM Alignment Revision	Normalized MSE: 0.310			25.8% zero-shot improvement via instruction steering				
Panel E: Cross-Modal Temporal Retrieval & Multi-Turn Agentic Reasoning										
Task / Benchmark	Model	Paradigm	R@1	R@5	MRR	Cross-Domain / OOD Generalization Finding				
Bidirectional TS-Text Retrieval (TRACE Benchmark) [27]	TS2Vec + BM25	Unimodal TS + Sparse Lexical	0.142	0.318	0.224	Unimodal embeddings fail semantic cross-modal alignment				
	Time-LLM (Zero-Shot) [7]	Reprogrammed Language Embeddings	0.285	0.542	0.391	Modest retrieval via generative prompt conditioning				
	TRACE [27]	Dual Contrastive Tower + Channel Tokens	0.518	0.784	0.627	+23.3% R@1 gain; robust zero-shot OOD retrieval across domains				
	InputAware-RAG [56]	Gated Cross-Attention Retrieval	0.492	0.761	0.605	Suppresses out-of-distribution noisy historical sequences				
Multi-Turn Agent Reasoning (TS-Agent Suite) [54]	Direct LLM Prompting TS-Agent	Frozen GPT-4o Zero-Shot Closed-Loop Tool-Augmented LLM	Success Rate: 34.2%			Hallucinates numerical trend values; lacks tool execution				
			Success Rate: 81.6%			+47.4% absolute reasoning gain via Python code/tool verification				
Panel F: Conformal Prediction Calibration, Continuous-Time State Space & Red-Teaming Robustness										
Task / Benchmark	Model	Paradigm	Coverage	MPiW	CRS	Safety, Efficiency & Resilience Finding				
Conformal Uncertainty Calibration (ETTth1 / Time-MMD, $1 - \alpha = 0.90$) [57], [58]	Standard Gaussian PI	Uncalibrated ($\pm z_{\alpha/2} \sigma$)	0.764	1.48	N/A	Severe undercoverage under heavy-tailed non-stationary shifts				
	Unimodal PatchTST + Split-CP	Unimodal Conformal	0.868	1.38	1.000	Nominal coverage deficit ($\sim 3.2\%$) due to unconditioned shifts				
	Time-LLM + Split-CP [57]	Multimodal Conformal Forecaster	0.901	1.02	0.420	Holds 90% coverage; 26.1% narrower MPiW via text context				
Wearable Physiological Imputation (PhysioNet / MIMIC-IV) [61]	Supervised Bi-LSTM SOTER [61]	Discrete Interpolation	Imputation RMSE: 0.218			Struggles with irregular multi-rate sensor missings				
		Continuous Neural CDE + MoE	Imputation RMSE: 0.142			34.9% RMSE reduction via continuous differential trajectories				
Adversarial Red-Teaming Audit (Synthetic Stress Harness) [62]	Time-LLM (Reprogrammed) ss-Manba / SOTER	Prompt-Conditioned Forecaster	0.754	1.02	0.420	Vulnerable to semantic inversion (SRR = 0.522); hallucinated ramps				
		Continuous-Time State Space	0.864	0.94	0.812	Highest resilience (CRS = 0.812); physical dynamics override text				
Panel G: Neuromorphic Energy Efficiency, Physics-Constrained Diffusion & Multi-Agent Swarms										
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding				
Micro-Watt Edge Forecasting (Energy / Power on ETT) [64], [66]	INT4-DeMa SSM TS-LIP [66]	Quantized Digital SSM	Energy: 2.10 mJ/token			Dense MAC arithmetic; 145 MB peak memory				
	SpikySpace [64]	Dual-Compartment SNN Spiking Selective SSM	Energy: 0.052 mJ/token			85x energy reduction vs INT4; < 65 mW power				
Physics-Constrained Simulation (Power Grid Swing SDE) [67], [68]	Unconstrained Diffusion	Multimodal Diff. Trans.	PDE Residual: 1.84×10^{-3}			Severe RoCoF violation; breaches IEEE stability limits				
	PhysDGM [68]	Stepwise Physics-Embedded	PDE Residual: 4.20×10^{-3}			97.7% PDE residual drop; enforces swing conservation				
Spatio-Temporal Swarm Reasoning (ST-Bench / Multi-Agent) [69], [70]	Single-Agent LLM	Zero-Shot CoT Prompting	Reasoning Acc: 54.2%			Hallucinates unfeasible spatial hops; lacks tool grounding				
	STReasoner / MAS4TS	S-GRPO Swarm Policy	Reasoning Acc: 88.5%			+34.3% reasoning accuracy; provable Byzantine resilience				
Panel H: Cross-Modal Causal Discovery, Microcontroller Distillation & Streaming Test-Time Adaptation										
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding				
Cross-Modal Causal Discovery (Macro-Financial CAMEF, $\gamma = 0.90$) [71]–[73]	Bivariate Granger / PCMCi+	Statistical Dependency	Edge F1: 16.8%–26.3%			Catastrophic breakdown under non-stationary unobserved confounding				
	MMoE Causal Guidance	MMoE Causal Guidance	Edge F1: 64.0% (MSE 0.384)			LLM future inferences bypass brittle representation-level alignment				
	CAMEF / Augur [71], [72]	Counterfactual M-SCM	Edge F1: 81.9% (MSE 0.372)			+55.4% F1 resilience; counterfactual augmentation bounds spurious discovery				
Microcontroller TSFM Distillation (ETTth1 / Weather, Cortex-M Limits) [74], [75]	Time-LLM Teacher [7]	7B Reprogrammed TSFM	Params: 7.0B, Memory: 14 GB			Unavailable for edge deployment ($> 250\text{W}$ power; 14 GB VRAM)				
	DistiITS [74]	Horizon-Weighted KD	Params: 4.8M, Memory: 410 KB			6000x speedup; fits < 512 KB SRAM and < 2 MB Flash ($\Delta\text{MSE} \leq 0.004$)				
	GUARD [75]	Gated Uncertainty KD	Params: 6.5M, Memory: 490 KB			Contextual router + circuit-breaker prevents negative transfer on science grids				
Streaming Test-Time Adaptation (Non-Stationary Shift on ETT) [76], [77]	Static Source Forecaster	Frozen Offline TSFM	Post-Shock MSE: 0.880			Severe error explosion (+130.4%) during abrupt regime transitions				
	Naive Gradient TTA	Standard Gradient Updates	Post-Shock MSE: 0.620			Catastrophic forgetting (34.2% loss of nominal baseline accuracy)				
	RG-TTA / TAFAS [76], [77]	Regime-Guided Meta-Control	Post-Shock MSE: 0.395			-52.1% shock MSE; W_1 + KS similarity meta-control eliminates forgetting				
Panel I: Neuro-Symbolic Logic Verification, Irregular Topologies & Federated Multimodal Adaptation										
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding				
Neuro-Symbolic Logic Verification (Waveform Event Detection, $d = 5$) [78], [79]	Direct LLM Prompting	Zero-Shot Prompting	Event F1: 24.0%			Fails under multi-channel temporal nesting; severe hallucinations				
	Standard VLM Chart QA	Direct Visual Inspection	Event F1: 38.1%			Plummet as logic depth grows; misses asynchronous channel triggers				
	Signal2Symbol [79]	Symbolic FOL Reasoner	Event F1: 78.2%			Discretized physiological state transitions enforce formal bounds				
	SELA (GrammarWave) [78]	Neuro-Symbolic Grammar	Event F1: 87.1%			+49.0% F1 surge vs VLMs; STL parse tree ensures 0 false invariants				
Irregular Spatio-Temporal Transfer (Climate / Traffic, 85% Missing) [80], [81]	Time Transformer	Zero-fill Patching	Forecasting MSE: 0.785			Degrades under asynchronous sampling; token context explodes				
	Spatio-Temporal GNN	Mean Imputation	Forecasting MSE: 0.790			Dyadic graph edges fail to represent high-order cross-sensor cliques				
	MHHyper-LLM [81]	Multi-Scale Hypergraph	Forecasting MSE: 0.560			Hyperedge incidence matrix H models non-pairwise correlations				
	LLMODE [80]	Neural ODE + Gated Token	Forecasting MSE: 0.410			-47.8% MSE drop; continuous-time dynamics robust to 85% missingness				
Privacy-Preserving Federated TSFM (Non-IID Drift $\alpha = 0.10$) [82]–[84]	Standard FedAvg	Full-Weight Sync	Multi-Client MSE: 0.465			Severe non-IID client drift; massive communication footprint (14 GB/round)				
	Naive Split Features	Incomplete Modality Distill	Multi-Client MSE: 0.410			Modality-invariant representations absorb heterogeneous patient sensors				
	FedChronos [82]	Federated PEFT LoRA	Multi-Client MSE: 0.388			Enforces strict data sovereignty and local differential privacy (98.5% less comm)				
	PerFed-TSFM [83]	Sparse Subnetwork Routing	Multi-Client MSE: 0.379			Converges in 20 rounds; decoupled sparse adaptation matches centralized accuracy				
Panel J: Embodied Robotics Telemetry, Quantum State Spaces, and Wireless Split Computing										
Task / Benchmark	Model	Paradigm	Primary Metric			Key Empirical Finding				
Bimanual Robotic Manipulation (Action Chunk Horizon $K = 24$) [85]–[87]	Single-Step BC	Autoregressive Step	Success Rate: 18.2%			Compounding error drift $O(T^2)$ causes catastrophic off-manifold failure				
	Fixed-Horizon MLP	Sliding Window MLP	Success Rate: 52.4%			Overfits to demonstration window; fails to handle contact discontinuities				
	ACT [85]	C-VAE + Transformer	Success Rate: 88.0%			Action chunking with temporal ensembling eliminates compounding initiation error				
	Diffusion Policy [86]	Continuous SDE Diffusion	Success Rate: 94.2%			Denoising trajectory diffusion captures multimodal action probability densities				
	HiPolicy [87]	Multi-Frequency Chunking	Success Rate: 96.5%			>38.3% over BC; decouples 2Hz semantic sub-goals from 50Hz joint control				
Long-Term Spatio-Temporal Prediction (Planetary / Traffic, $H = 1080$) [88], [89]	Informor / Autoformer	Classical Attention	Forecasting MSE: 0.760			Attention degradation and error accumulation across multi-day horizons				
	Mamba SG	Classical State Space	Forecasting MSE: 0.582			Linear recurrence suffers hidden-state drift under extreme non-stationarity				
	H-STQGCN [89]	Quantum Graph ConvNet	Forecasting MSE: 0.465			Entanglement gates model non-local topological correlations across arbitrary hops				
	Quantum-Mamba [88]	PQC + Selective SSM	Forecasting MSE: 0.388			-48.9% MSE vs Transformers; unitary norm preservation prevents gradient decay				
Wireless Split Computing Inference (40% Packet Loss Rate $\eta = 0.40$) [90], [91]	Raw Sensor Streaming	Full Waveform TCP/UDP	Task Accuracy: 14.2%			High channel loss corrupts time-domain waveforms; latency exceeds 150 ms				
	Naive Split Features	Dense Hidden Vectors	Task Accuracy: 32.5%			Dropped intermediate features diminish stateful classification manifolds				
	Resonate-and-Fire Split [90]	RF Spiking Neurons	Task Accuracy: 85.0%			Oscillatory resonance preserves frequency peak semantics under sub-millisecond power				
	Semantic TS Autoenc. [91]	Goal-Oriented Data-Distort.	Task Accuracy: 94.2%			12.8x compression; goal-oriented coding preserves semantics under 40% loss				